

RESEARCH ARTICLE

Robust Adaptive H_∞ Tracking Control for Nonlinear Strict-Feedback System via Actor-Critic-Disturbance Learning: A Zero-Sum Game-Based Approach

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ABSTRACT

This article presents a robust adaptive H_∞ tracking control scheme for a class of nonlinear strict-feedback systems. Building upon robust control principles for disturbance attenuation and the nonlinear backstepping technique for stabilization, the proposed approach unifies robust H_∞ tracking design with online adaptive dynamic programming for strict-feedback nonlinear systems. Finite-energy disturbances are explicitly modeled as an adversarial player in a zero-sum game, thereby bridging the gap between robust tracking synthesis and data-driven online optimization. To approximate the Nash equilibrium, a novel Actor-Critic-Disturbance learning algorithm is developed, where the sufficiency of excitation condition combined with the experience replay mechanism ensures adaptive weight convergence without relying on the restrictive persistence of excitation. Furthermore, the value function is decomposed into a quadratic part and a nonlinear component, thereby guaranteeing satisfactory tracking performance while avoiding direct solutions of the Hamilton–Jacobi–Isaacs equation. Theoretical analysis establishes rigorous justification for the stability of the overall closed-loop system, and simulation studies validate the effectiveness of the proposed method.

1 | Introduction

Robust control has long played a central role in addressing system uncertainties and external disturbances [1], while nonlinear control methods provide fundamental tools for stabilizing complex dynamical systems [2]. Among them, the backstepping technique [3] has emerged as a powerful recursive design technique that leverages Lyapunov functions and virtual control signals to handle strict-feedback systems [4, 5]. In parallel, adaptive dynamic programming (ADP) has been widely explored to approximate optimal control laws in uncertain settings [6–10]. More recently, game-theoretic methods [11], particularly zero-sum games (ZSGs), have provided a natural way to

characterize the interaction between control input and adversarial disturbance in H_∞ control problems [12–14]. Motivated by these developments, this article integrates ADP with backstepping within a ZSG formulation to address the robust H_∞ tracking control problem for a class of nonlinear strict-feedback systems.

1.1 | Related work

Over the past decades, the theories of robust control and nonlinear backstepping design have undergone rapid development, with significant contributions both in methodology [15] and in engineering applications [16]. In robust control, H_∞ techniques

have been extensively investigated for disturbance attenuation and uncertainty compensation in nonlinear systems [17–19]. Meanwhile, backstepping has been recognized as one of the most versatile methods for strict-feedback systems, providing a constructive procedure for controller synthesis through recursive Lyapunov designs. This approach has been widely applied to robotics [20], flight control [21], and motors [22], to name a few. This demonstrates its ability to stabilize nonlinear dynamics under parameter variations and external disturbances. Moreover, extensions such as dynamic surface control (DSC) have been developed to alleviate the “explosion of complexity” problem in backstepping, making the methodology more practical for high-order systems [23]. These advances firmly establish backstepping as a central tool in nonlinear robust control.

In parallel, the integration of backstepping with ADP methods has attracted growing attention, as ADP provides the capability of approximating optimal feedback policies in an online manner [24]. Actor-Critic structures in ADP allow the approximation of the Hamilton–Jacobi–Bellman solution, enabling real-time optimal control [6, 25]. When combined with backstepping, these approaches have been successfully employed in strict-feedback nonlinear systems to achieve adaptive optimal tracking [26, 27]. However, many existing studies still adopt a relatively simplified treatment of disturbances, typically by assuming that they are bounded exogenous signals and then mitigating their influence either through observer design or by resorting to conservative mathematical relaxations [28]. While such approaches may offer analytical tractability, they inevitably lead to restrictive controller designs and overlook more realistic disturbance patterns. In numerous practical scenarios—such as robotic-arm motion planning subject to transient impacts and friction uncertainties [29], wireless-powered cooperative communication networks with bursty interference and imperfect energy harvesting [30], and cyber–physical systems experiencing sporadic cyber/communication-induced perturbations [31]—disturbances are more appropriately characterized as finite-energy $\mathcal{L}_2$ signals rather than pointwise bounded inputs [32]. This modeling choice is physically meaningful for mechanical and power systems, where disturbances often arise from short-duration transients (e.g., load steps, switching events, impulses) whose instantaneous magnitude may be large but whose accumulated energy over time remains limited. Moreover, the finite-energy description yields a less restrictive and more general formulation than the bounded-disturbance setting, while naturally aligning with the $H_\infty/\mathcal{L}_2$ consideration. In practice, the disturbance itself is typically not directly measurable. Only corrupted state/output measurements are available, and the “energy” of the disturbance must be inferred from data, which can be challenging due to sensor noise, limited bandwidth, sampling, and unmodeled dynamics. This naturally motivates the adoption of a ZSG framework, within which the controller is designed to achieve robustness against the worst-case disturbance input, thereby laying a rigorous foundation for robust adaptive control under energy-constrained perturbations.

Nonetheless, a major difficulty in ADP-based backstepping tracking designs lies in handling the value function, since directly approximating it often complicates stability analysis and demands heavy function approximation. To address this, the

value function can be separated into a quadratic term and a nonlinear residual component: the quadratic part inherently contributes to ensuring satisfactory tracking performance, while only the residual term is approximated by neural networks [24]. This treatment not only simplifies the stability proof but also circumvents the need to solve the Hamilton–Jacobi–Isaacs equation (HJIE) [11, 33] directly. In parallel, another challenge of existing methods is the reliance on the persistence of excitation (PE) condition [34] for weight convergence during the online learning process [6]. This requirement is often impractical, as it demands continuous excitation over an infinite horizon and may contradict the smooth operation of engineering systems [35]. To mitigate this issue, the experience replay mechanism is adopted, allowing historical data to be reused concurrently with online samples, thereby relaxing the restrictive PE requirement with a weaker sufficiency of excitation (SE) criterion while still ensuring stable adaptation [35–37].

Contributions: Collectively, this article develops a robust adaptive H_∞ tracking control framework for nonlinear strict-feedback systems by a ZSG reformulation of backstepping-based Actor-Critic-Disturbance algorithm. The main contributions of this article are threefold:

1. Existing robust H_∞ –ADP studies typically treat the disturbance as a bounded exogenous signal [28] or attempt to reconstruct it through disturbance observers [38], which can add estimation burden and may mismatch energy-limited perturbations. In contrast, we model the disturbance as an adversarial player in a ZSG formulation, directly embedding finite-energy effects into the control framework for energy-aware robust tracking.
2. Conventional backstepping for strict-feedback systems may suffer from unknown disturbances and lengthy Lyapunov-based derivations [26, 39]. In contrast, we cast the Actor-Critic-Disturbance backstepping design into a two-player ZSG, offering a clear game-theoretic interpretation for robust H_∞ tracking and yielding an approximate Nash-equilibrium solution without tedious Lyapunov-based stability proofs.
3. Many ADP-based backstepping approaches require PE for convergence [24, 40], which is restrictive in practice and can lead to performance degradation. In contrast, we incorporate experience replay under an SE condition to enable stable online learning without PE. Moreover, the value-function decomposition (quadratic and nonlinear remainder) further avoids directly solving the HJIE and reduces the burden of full value estimation while the quadratic component inherently preserves tracking performance.

Structure: The remainder of this article is arranged as follows. Section 2 offers preliminaries of the H_∞ control and game theory together with the function approximation techniques. In Section 3, we state the tracking control problem of the strict-feedback system. In Section 4, we first transform the tracking control problem into the solvability of the HJIE. Then, a novel Actor-Critic-Disturbance composite learning algorithm is developed through experience replay scheme, with theoretical

justifications. Simulation studies in Section 5 verify the robustness of the proposed learning method, followed by conclusions in Section 6.

2 | Preliminaries

2.1 | Finite-Gain $\mathcal{L}_2$ Stability

Consider the smooth nonlinear system in the input-output form of:

$$\begin{aligned}\dot{x} &= f_d(x) + k_d(x) d, \\ y_d &= h_d(x),\end{aligned}\quad (1)$$

where $x(t) \in \mathbb{R}^{n_x}$ is the system state with the initial value $x(0) = x_0$, $d(t) \in \mathbb{R}^{n_d}$ denotes the disturbance input, and $y_d(t) \in \mathbb{R}^{n_y}$ is defined as the fictitious performance output. The functions $f_d : \mathbb{R}^{n_x} \mapsto \mathbb{R}^{n_x}$ and $k_d : \mathbb{R}^{n_x} \mapsto \mathbb{R}^{n_x \times n_d}$ are locally Lipschitz continuous, representing the drift and disturbance dynamics, respectively. Without loss of generality, we assume that f_d and h_d vanish at the origin [41], that is, $f_d(0) = 0$ and $h_d(0) = 0$.

For subsequent discussions, the relevant definitions and lemmas are provided.

Definition 1 (Zero-State Observable [42]). The system described by (1) is said to be locally zero-state observable if there exists a neighborhood $\mathcal{O}$ of the origin, such that:

$$d(t) \equiv 0 \quad \text{and} \quad y_d(t) \equiv 0 \quad \Rightarrow \quad x(t) \equiv 0, \quad \forall x \in \mathcal{O}. \quad (2)$$

If $\mathcal{O} = \mathbb{R}^{n_x}$, the system (1) is zero-state observable (ZSO).

Definition 2 (Available Storage [43]). With supply rate $r(y, d)$, the available storage V_s of the system (1) is defined as

$$V_s(x_0) \triangleq \sup_{d \in \mathcal{L}_2, T > 0} \int_0^T -r(y_d(t), d(t)) dt. \quad (3)$$

The storage function defined above can be regarded as the maximal extractable energy from the input-output system (1). Combined with the ZSO condition (2), the available storage can be applied to the Lyapunov stability analysis of system (1), as detailed in the lemma below.

Lemma 1 (Relation between ZSO and available storage [32]). *Select the supply rate in (3) as $r(y_d, d) = \gamma^2 \|d\|^2 - \|y_d\|^2$ with $\gamma \geq 0$. Let the available storage V_s in (3) be positive semi-definite and satisfy the dissipative inequality $\dot{V}_s(x(t)) \leq r(y_d(t), d(t))$. If the system (1) is ZSO, the following conclusions hold:*

- The unperturbed system $\dot{x} = f_d(x)$ is asymptotically stable at the origin, and V_s is positive definite.
- The unperturbed system $\dot{x} = f_d(x)$ is globally asymptotically stable at the origin, provided that V_s is radially unbounded.

In what follows, we introduce the definition of $\mathcal{L}_2$ -gain.

Definition 3 ($\mathcal{L}_2$ -Gain [17]). With $\gamma \geq 0$, the system (1) is said to have $\mathcal{L}_2$ -gain less than or equal to γ , provided that

$$\int_0^t \|y_d(\tau)\|^2 d\tau \leq \gamma^2 \int_0^t \|d(\tau)\|^2 d\tau \quad \forall t \geq 0 \quad \text{and} \quad \forall d \in \mathcal{L}_2. \quad (4)$$

Definition 4 (Finite-gain $\mathcal{L}_2$ stability [2]). The system (1) is finite-gain $\mathcal{L}_2$ stable, if and only if $\exists \gamma, m \in \mathbb{R}^+ \cup \{0\}$ satisfying

$$\|y_d\|_{\mathcal{L}_2} \leq \gamma \|d\|_{\mathcal{L}_2} + m, \quad \forall d \in \mathcal{L}_2. \quad (5)$$

The following lemma bridges the gap between Lyapunov stability and finite-gain $\mathcal{L}_2$ stability.

Lemma 2. [2] *Suppose the system (1) is exponentially stable at the origin. With some positive constants p_1, p_2, p_3, p_4 , and p_x , let $V \in C^1 : \mathbb{R}^{n_x} \mapsto \mathbb{R}$ satisfy: (i) $p_1 \|x\|^2 \leq V(x) \leq p_2 \|x\|^2$, (ii) $\nabla^T V(x) f_d(x) \leq -p_3 \|x\|^2$, (iii) $\|\nabla V(x)\| \leq p_4 \|x\|$, for $\|x\| \leq p_x$. Then, system (1) is finite-gain $\mathcal{L}_2$ stable. That is, $\forall d \in \mathcal{L}_2$, (5) holds with $x_0 \leq p_x \sqrt{p_1/p_2}$ and sufficiently small $\|d\|_{\mathcal{L}_\infty}$.*

Remark 1. The signal y_d is a user-defined performance output introduced for finite-gain $\mathcal{L}_2$ analysis, which need not coincide with a sensor output. It aggregates the quantities whose energy we wish to penalize (e.g., tracking errors and/or control effort) so that dissipation inequalities can express disturbance attenuation.

2.2 | Zero-Sum Game Theory

Denote u and d the minimizing and maximizing players, respectively, of the 2-player ZSG [12]:

$$\min_{u \in \Omega_u} \max_{d \in \mathcal{L}_2} J(u, d, \gamma; x_0), \quad (6)$$

$$\text{s.t.} \quad \begin{cases} \dot{x} = f(x) + g(x)u + k(x)d, & x(0) = x_0, \\ y = h(x, u) \end{cases} \quad (7)$$

where the long-term performance index (target of the two players u and d) of the system (7) is given by

$$J(u, d, \gamma; x_0) \triangleq \int_t^\infty l(x(\tau), u(\tau), d(\tau)) d\tau.$$

with the instantaneous cost $l(x, u, d) \triangleq \|h(x, u)\|^2 - \gamma^2 \|d\|^2$. (7) is the system of interest, with the system state $x(t) \in \mathbb{R}^n$ and its initial value x_0 , and the disturbance input $d(t) \in \mathbb{R}^{n_d}$. Locally Lipschitz continuous functions $f : \mathbb{R}^n \mapsto \mathbb{R}^n$, $g : \mathbb{R}^n \mapsto \mathbb{R}^{n \times n_u}$, $k : \mathbb{R}^n \mapsto \mathbb{R}^{n \times n_d}$ stand for the drift, input, and disturbance dynamics, respectively, with $f(0) = 0$. $y(t) \in \mathbb{R}$ is the instantaneous performance output at time t given by $y = h(x, u) = Q(x) + u^T R u$ with positive definite function $Q(x)$ and positive definite symmetric matrix $R \in \mathbb{R}^{n_u \times n_u}$.

For system (7), define a set Ω_u that contains all locally Lipschitz continuous feedback policies u such that:

- For $d(t) \equiv 0$, the system (7) is asymptotically stable at the origin;
- For all $d(t) \in \mathcal{L}_2$, the system (7) is finite-gain $\mathcal{L}_2$ stable;
- $u = u(x)$ with $u(0) = 0$.

Then, consider the disturbance attenuation problem [17]. Specifically, for the given $\gamma \geq \gamma^* \geq 0$ with γ^* being the smallest attenuation level, one needs to find the control input $u \in \Omega_u$ such that

$$\sup_{d \in \mathcal{L}_2} \frac{\|h(x, u)\|_{\mathcal{L}_2}}{\|d\|_{\mathcal{L}_2}} \leq \gamma. \quad (8)$$

Based on the dynamic game theory [12], achieving the disturbance attenuation (8) is equivalent to solving the two-player ZSG (6). To begin with, define the value function corresponding to system (7) as

$$\mathcal{P}(x; u, d) \triangleq \int_t^\infty l(x(\tau), u(\tau), d(\tau)) d\tau. \quad (9)$$

Then, we give the following definition, which is closely related to the ZSG (6).

Definition 5 (Nash equilibrium [12]). An input pair (u^*, d^*) is said to constitute a Nash equilibrium to the ZSG (6), if and only if:

$$\mathcal{P}(x; u^*, d) \leq \mathcal{P}(x; u^*, d^*) \leq \mathcal{P}(x; u, d^*). \quad (10)$$

When the saddle point condition [44] holds:

$$\min_{u \in \Omega_u} \max_{d \in \mathcal{L}_2} \mathcal{P}(x; u, d) = \max_{d \in \mathcal{L}_2} \min_{u \in \Omega_u} \mathcal{P}(x; u, d), \quad (11)$$

one can obtain the unique Nash equilibrium solution to the ZSG (6) in the form of

$$u^*(x) = -\frac{1}{2} R^{-1} g^\top \nabla \mathcal{P}^*(x), \quad d^*(x) = \frac{1}{2\gamma^2} k^\top \nabla \mathcal{P}^*(x), \quad (12)$$

where $\mathcal{P}^*(x) \triangleq \mathcal{P}(x; u^*(x), d^*(x)) = \min_{u \in \Omega_u} \max_{d \in \mathcal{L}_2} \mathcal{P}(x; u, d) = \max_{d \in \mathcal{L}_2} \min_{u \in \Omega_u} \mathcal{P}(x; u, d)$, is called the *value* of the ZSG (6). Thus, to find the Nash solution (12) is to find the value function $\mathcal{P}^*(s)$, which can be obtained by solving the HJIE:

$$\mathcal{H}(x, u^*(x), d^*(x), \nabla \mathcal{P}^*(x)) = 0, \quad \mathcal{P}^*(0) = 0, \quad (13)$$

where $\mathcal{H}(x, u^*(x), d^*(x), \nabla \mathcal{P}^*(x)) \triangleq l(x, u^*(x), d^*(x)) + \nabla^\top \mathcal{P}^*(x)[f(x) + g(x)u^*(x) + k(x)d^*(x)]$.

Suppose the solution $\mathcal{P}^*(x)$ to (13) is positive semi-definite [45]. Therefore, by Lemma 1 and the ZSO condition (2), one has: (i) $\mathcal{P}^*(x)$ is positive definite; (ii) The system (7) is asymptotically stable under $u^*(x)$ at the origin for $d \equiv 0$; (iii) The system (7) is globally asymptotically stable under $u^*(x)$ at the origin for $d \equiv 0$ if $\mathcal{P}^*(x)$ is radially unbounded additionally.

2.3 | Function Approximation

In order to cope with unknown nonlinear functions, function approximation frameworks are widely employed in the field of nonlinear control, where common choices of basis functions include polynomial basis function (PBF), radial basis function (RBF), fuzzy basis function, and so forth. Throughout this article, we implement the RBF neural network (RBFNN). On a compact

set $\Omega_\chi \subset \mathbb{R}^n$, the RBFNN representation of an unknown continuous function $M : \Omega_\chi \mapsto \mathbb{R}$, is given by

$$\hat{M}(\chi) = \theta^\top \varphi(\chi). \quad (14)$$

In the approximation (14), $\theta \in \mathbb{R}^N$ is the weight vector; $\varphi : \mathbb{R}^n \mapsto \mathbb{R}^N$ is the RBFNN activation function vector:

$$\varphi(\chi) = [\varphi_1(\chi), \dots, \varphi_N(\chi)]^\top, \quad \varphi_i(\chi) = \exp\left(-\frac{\|\chi - \mu_i\|^2}{2\sigma_i^2}\right) \in \mathbb{R}, \quad (15)$$

$i = 1, 2, \dots, N,$

with $\mu_i \in \Omega_\chi$ denoting the centers and $\sigma_i \in \mathbb{R}$ being the widths of the Gaussian functions. The universal approximation property of the RBFNN can be established as follows.

Lemma 3 (Universal Approximation [46]). *On the compact set Ω_χ , consider an arbitrary continuous mapping $M : \Omega_\chi \mapsto \mathbb{R}$. Then, there exists an RBFNN in the form of (14), such that for given accuracy $\varepsilon > 0$ and a sufficiently large number of hidden nodes N , the statements listed below hold:*

i. *The approximation error $\varepsilon^*(\chi) \triangleq M(\chi) - \theta^{*\top} \varphi(\chi)$ is uniformly bounded on Ω_χ , such that $|\varepsilon^*(\chi)| \leq \varepsilon$, where*

$$\theta^* = \arg \min_{\theta} \left\{ \sup_{\chi} |M(\chi) - \theta^\top \varphi(\chi)| \right\}. \quad (16)$$

ii. *As the number of nodes $N \rightarrow \infty$, $\varepsilon^* \rightarrow 0$ and $\nabla \varepsilon^* \rightarrow 0$ uniformly.*

iii. *For fixed N , the approximation error ε^* and its gradient $\nabla \varepsilon^*$ are bounded by positive constants on the compact set Ω_χ .*

3 | Problem Statement

In this article, we consider the following nonlinear strict-feedback system in the form of:

$$\begin{aligned} \dot{x}_i &= f_i(\bar{x}_i) + g_i(\bar{x}_i)x_{i+1} + k_i(\bar{x}_i)d_i, \quad i = 1, \dots, n-1, \\ \dot{x}_n &= f_n(\bar{x}_n) + g_n(\bar{x}_n)u + k_n(\bar{x}_n)d_n, \end{aligned} \quad (17)$$

where $\bar{x}_i = [x_1, \dots, x_i]^\top \in \mathbb{R}^i$ is the system state; $u \in \mathbb{R}$ the control input; $d \in \mathbb{R}$ the disturbance input with finite energy, that is, $d_i \in \mathcal{L}_2$; $x_1 \in \mathbb{R}$ the system output. The known functions f_i , g_i , and k_i are locally Lipschitz continuous on compact sets Ω_i for $i = 1, \dots, n$. In this article, we consider the tracking problem.

Let $\delta_0(t) = x_d(t)$, and δ_{i-1} for $i = 2, \dots, n$ be the DSC internal filter states that will be specified later. Let δ_{i-1} for $i = 1, \dots, n$ be the reference signals to be tracked by each state x_i , $i = 1, \dots, n$. Accordingly, define the following coordination transformation:

$$z_i = x_i - \delta_{i-1} \quad \text{for } i = 1, \dots, n. \quad (18)$$

Assumption 1. [17] For the bounded reference trajectories δ_{i-1} defined on the compact set Ω_i , there exist corresponding bounded Lipschitz continuous command generator functions $\ell_{i-1}(\cdot) \in \mathbb{R}$ with $\ell_{i-1}(0) = 0$ such that for $i = 1, \dots, n$ and $\bar{\ell} \in \mathbb{R}^+$:

$$\dot{\delta}_{i-1}(t) = \ell_{i-1}(\delta_{i-1}(t)) \quad \text{and} \quad \|\ell_{i-1}(\cdot)\| \leq \bar{\ell}. \quad (19)$$

Then, we introduce the DSC filter [23] with user-defined time parameter $\tau_i > 0$:

$$\tau_i \dot{\delta}_i = u_i - \delta_i \quad \text{for } i = 1, \dots, n-1, \quad (20)$$

where u_i are the virtual control inputs act on the first $n-1$ subsystems, and the corresponding filter errors are given by

$$\tilde{\delta}_i = \delta_i - u_i. \quad (21)$$

With (18) and (21), the strict-feedback system (17) can be rewritten as

$$\begin{aligned} \dot{x}_i &= f_i(\bar{x}_i) + g_i(\bar{x}_i)u_i + k_i(\bar{x}_i)d_i + \zeta_i, \quad i = 1, \dots, n-1, \\ \dot{x}_n &= f_n(\bar{x}_n) + g_n(\bar{x}_n)u_n + k_n(\bar{x}_n)d_n, \end{aligned} \quad (22)$$

where $u_n \triangleq u$ for consistency and $\zeta_i \triangleq g_i \cdot (z_{i+1} + \tilde{\delta}_i)$.

Remark 2. (19) describes the command generator enforced on the bounded reference trajectory $\delta_{i-1}(t)$ for the coordinate transformation and augmented system construction (with $\delta_0 \triangleq x_d$), $i = 1, 2, \dots, n$. (20) introduces the DSC filter whose state δ_i is driven by the virtual control input u_i (for $i = 1, \dots, n-1$) to avoid repeated differentiation of virtual controls. Hence, (19) characterizes the evolution of the reference trajectory, whereas (20) provides an implementable dynamic filtering mechanism for realizing the virtual controllers, while leaving the command generator representation in (19) unchanged.

Problem 1. Consider the system described in (17), one control objective is to design the real control input u and the virtual controllers u_i such that the system output $x_1(t)$ is driven to track the given reference signal $x_d(t)$ asymptotically.

On the compact sets Ω_i , it is assumed there exist $\bar{f}, \bar{g}, \bar{g}, \bar{k} \in \mathbb{R}^+$, such that functions f_i, g_i , and k_i are bounded as:

$$|f_i(\cdot)| \leq \bar{f}, \quad \bar{g} \leq |g_i(\cdot)| \leq \bar{g}, \quad |k_i(\cdot)| \leq \bar{k}. \quad (23)$$

Based on (18) and Assumption 1, the dynamics of the tracking error can be expressed as

$$\begin{aligned} \dot{z}_i &= f_i + g_i u_i + k_i d_i + \zeta_i - \ell_i(\delta_{i-1}). \\ \dot{z}_n &= f_n + g_n u_n + k_n d_n - \ell_i(\delta_{n-1}). \end{aligned} \quad (24)$$

Since the error dynamics also contains the dynamics of the trajectory α_{i-1} , define the augmented state for the i th subsystem as

$$\mathcal{X}_i \triangleq [z_i, \delta_{i-1}]^\top \in \mathbb{S}_i \subset \mathbb{R}^2, \quad (25)$$

where $\mathbb{S}_i$ is a compact set containing the origin. Then, for each i , one can formulate the following augmented subsystem:

$$\dot{\mathcal{X}}_i = \mathcal{F}_i + \mathcal{G}_i u_i + \mathcal{K}_i d_i + \Delta_i, \quad (26)$$

where $\Delta_i = [\zeta_i, 0]^\top$ for $i = 1, \dots, n-1$ and $\Delta_n = \mathbf{0}_{2 \times 1}$; $\mathcal{F}_i \triangleq [f_i - \ell_i, \ell_i]^\top$, $\mathcal{G}_i \triangleq [g_i, 0]^\top$, $\mathcal{K}_i \triangleq [k_i, 0]^\top$ represent the drift, input, disturbance dynamics of the augmented subsystem, respectively.

In (26), the drift term $\mathcal{F}_i$ stands for $\mathcal{F}_i(\bar{\mathcal{X}}_i)$ with $\bar{\mathcal{X}}_i \triangleq [\mathcal{X}_1, \dots, \mathcal{X}_i]^\top$ since $\mathcal{F}_i$ contains $f_i(\bar{x}_i)$. In the strict-feedback backstepping/DSC design, the closed-loop system has a cascaded structure, and by tuning the DSC time constants and feedback gains (which will be specified later), the inner-loop variables (states closer to the actual control input) can be made to converge faster than the outer-loop variables (states far away the actual control input). Therefore, at the design step i , $\bar{\mathcal{X}}_{i-1}(t)$ is regarded as a slowly-varying/constant. For notational simplicity, this dependence on $\bar{\mathcal{X}}_{i-1}$ is omitted in $\mathcal{F}_i$.

4 | Main Result

4.1 | H_∞ Tracking Control and Zero-Sum Game Formulation

In this subsection, the H_∞ tracking control problem of each augmented system (26) is formulated through a general $\mathcal{L}_2$ -gain condition with a discounted performance index [47], as discussed in the following.

For each augmented subsystem (26), define the fictitious performance output $\omega_i(\mathcal{X}_i, u_i)$ as

$$\omega_i(\mathcal{X}_i, u_i) \triangleq \sqrt{\mathcal{X}_i^\top Q_i \mathcal{X}_i + r_i u_i^2}, \quad (27)$$

where $Q_i \triangleq \text{diag}(q_i, 0)$ with $q_i > 0$, and $r_i > 0$. Accordingly, we introduce the discounted performance index

$$J_i(u_i, d_i, \gamma_i; \mathcal{X}_i) \triangleq \int_t^\infty e^{-\eta(\tau-t)} \rho_i(\mathcal{X}_i(\tau), u_i(\tau), d_i(\tau)) d\tau, \quad (28)$$

where $\rho_i(\mathcal{X}_i, u_i, d_i) \triangleq \omega_i^2(\mathcal{X}_i, u_i) - \gamma_i^2 d_i^2$ is the instantaneous cost, $\eta > 0$ is the discount factor, and γ_i is the disturbance attenuation level. Respectively, the performance output and index of the overall error dynamics (24) are defined as

$$\omega(\bar{\mathcal{X}}, \bar{u}) \triangleq \sqrt{\sum_{i=1}^n \omega_i^2(\mathcal{X}_i, u_i)}, \quad (29)$$

$$J(\bar{u}, \bar{d}, \bar{\gamma}; \bar{\mathcal{X}}) \triangleq \sum_{i=1}^n J_i(u_i, d_i, \gamma_i; \mathcal{X}_i), \quad (30)$$

where $\bar{u} \triangleq [u_1, \dots, u_n]^\top$, $\bar{d} \triangleq [d_1, \dots, d_n]^\top$, $\bar{\mathcal{X}} \triangleq [\mathcal{X}_1^\top, \dots, \mathcal{X}_n^\top]^\top$, and $\bar{\gamma} \triangleq \max_i \{\gamma_i\}$. Then, another control objective is to attenuate the disturbance input $\bar{d}$, such that the following general $\mathcal{L}_2$ -gain condition is satisfied [17, 48, 49]:

$$\sup_{d \in \mathcal{L}_2} \left[\frac{\int_t^\infty e^{-\eta(\tau-t)} \omega^2(\bar{\mathcal{X}}(\tau), \bar{u}(\tau)) d\tau}{\int_t^\infty e^{-\eta(\tau-t)} \|\bar{d}(\tau)\|^2 d\tau} \right]^{\frac{1}{2}} \leq \bar{\gamma}. \quad (31)$$

Remark 3. Different from regulation problems, this work addresses an infinite-horizon tracking objective. Hence, even when the tracking error converges to a neighborhood of the origin, the steady-state control input u_i (and the induced closed-loop signals) does not necessarily decay to the origin. In this case, an undiscounted energy-based attenuation condition may lead to an unbounded (or ill-defined) performance function (28). To ensure

the well-posedness of the attenuation measure and the boundedness of the associated value function in the tracking setting, we adopt the discounted form in (31) by introducing the exponential weight $e^{-\eta(\tau-t)}$ with $\eta > 0$.

Remark 4. The discount factor η serves as a design parameter balancing long-horizon performance emphasis and integrability: smaller η allows smaller tracking errors which will be discussed in subsequent sections, whereas larger η improves boundedness of the value function.

Then, Problem 1 can be extended to the following problem.

Problem 2 (H_∞ tracking control). For each augmented subsystem (26), the H_∞ tracking control problem is to find the control input $u_i \in \mathbb{U}$ for $i = 1, \dots, n$, where $\mathbb{U}$ is a set of all locally Lipschitz continuous feedback policies, such that

- $u_i = u_i(\mathcal{X}_i)$ with $u_i(0) = 0$;
- The system (24) is locally asymptotically stable at the origin for $d_i(t) \equiv 0$;
- The system (24) is finite-gain $\mathcal{L}_2$ stable for all $d_i(t) \in \mathcal{L}_2$, that is, the generalized $\mathcal{L}_2$ -gain condition (31) is satisfied.

Define the discounted value function for each subsystem as

$$\mathcal{V}_i(\mathcal{X}_i; u_i, d_i) = \int_t^\infty e^{-\eta(\tau-t)} \rho_i(\mathcal{X}_i(\tau), u_i(\tau), d_i(\tau)) d\tau. \quad (32)$$

Following the discussions in Section 2.2, Problem 2 can be interpreted into the ZSG in the form of:

$$\min_{u_i \in \mathbb{U}} \max_{d_i \in \mathcal{L}_2} \mathcal{V}_i(\mathcal{X}_i; u_i, d_i) \quad (33)$$

with the saddle point condition [12]:

$$\min_{u_i \in \mathbb{U}} \max_{d_i \in \mathcal{L}_2} \mathcal{V}_i(\mathcal{X}_i; u_i, d_i) = \max_{d_i \in \mathcal{L}_2} \min_{u_i \in \mathbb{U}} \mathcal{V}_i(\mathcal{X}_i; u_i, d_i). \quad (34)$$

Then, the Nash equilibrium $\{u_i^*, d_i^*\}$ is constituted of

$$u_i^*(\mathcal{X}_i) = -\frac{1}{2r_i} \mathcal{G}_i^\top \nabla \mathcal{V}_i^*(\mathcal{X}_i), \quad (35)$$

$$d_i^*(\mathcal{X}_i) = \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla \mathcal{V}_i^*(\mathcal{X}_i), \quad (36)$$

where $\mathcal{V}_i^*(\mathcal{X}_i) \triangleq \mathcal{V}_i(\mathcal{X}_i; u_i^*(\mathcal{X}_i), d_i^*(\mathcal{X}_i))$ is the value of the ZSG (33), and uniquely solves the following HJIE:

$$\begin{aligned} 0 = & \mathcal{H}_i(\mathcal{X}_i, u_i^*(\mathcal{X}_i), d_i^*(\mathcal{X}_i), \nabla \mathcal{V}_i^*(\mathcal{X}_i)) \triangleq \rho_i(\mathcal{X}_i, u_i^*(\mathcal{X}_i), d_i^*(\mathcal{X}_i)) \\ & + \nabla^\top \mathcal{V}_i^*(\mathcal{X}_i) \cdot [\mathcal{F}_i + \mathcal{G}_i u_i^*(\mathcal{X}_i) + \mathcal{K}_i d_i^*(\mathcal{X}_i) + \Delta_i] \\ & - \eta \mathcal{V}_i^*(\mathcal{X}_i). \end{aligned} \quad (37)$$

Although $\mathcal{F}_i$ in the HJIE (37) contains $\bar{\mathcal{X}}_i$ (and thus includes the preceding states), the HJIE at step i is solved with respect to the local augmented state $\mathcal{X}_i$. This is because the backstepping/DSC design induces a time-scale separation in the cascaded strict-feedback chain: once the inner-loop tracking/filter

dynamics becomes sufficiently fast, the preceding states $\bar{\mathcal{X}}_{i-1}(t)$ can be treated as bounded slowly-varying signals or constants when constructing the step- i value function. Hence, $\mathcal{V}_i^*$ is expressed as a function of $\mathcal{X}_i$ (more precisely, $\mathcal{V}_i^*(\mathcal{X}_i; \bar{\mathcal{X}}_{i-1})$), and the dependence on $\bar{\mathcal{X}}_{i-1}$ is suppressed for brevity. Similar treatments are standard in strict-feedback DSC/backstepping and ADP-based tracking designs [24, 26, 40, 50].

Remark 5. In the presented robust H_∞ setting, the Nash equilibrium is a saddle point such that: the control policy u_i^* minimizes the discounted performance under the worst admissible finite-energy disturbance, while d_i^* maximizes the same performance under the best admissible controller. The saddle-point equality (34) formalizes that neither side can unilaterally reduce/increase the value once $\{u_i^*, d_i^*\}$ is reached, which is the game-theoretic counterpart of worst-case disturbance attenuation [12, 32].

In this article, we assume each augmented subsystem (26) under consideration satisfy the ZSO condition [51] and the HJIE (37) uniquely admits a positive semi-definite solution [32]. Then, by Lemma 1, $\mathcal{V}_i^*(\mathcal{X}_i)$ is positive definite. Hence, for stability analysis, we choose $\mathcal{V}_i^*(\mathcal{X}_i)$ as the Lyapunov candidate for each subsystem, and define

$$\mathcal{V}_\Sigma^*(\bar{s}) \triangleq \sum_i \mathcal{V}_i^*(\mathcal{X}_i) \quad (38)$$

as the Lyapunov function of the whole dynamical system (24). Additionally, the following standard assumption is adopted.

Assumption 2. [52] For $i = 1, \dots, n$, there exists $\eta^* \in \mathbb{R}^+ > \eta$ satisfying $\eta^* \mathcal{V}_i^*(\mathcal{X}_i) \leq \rho_i(\mathcal{X}_i, u_i^*(\mathcal{X}_i), d_i^*(\mathcal{X}_i))$, $\forall \mathcal{X}_i \in \mathbb{S}_i$.

Assumption 2 requires the stage cost $\rho_i(\mathcal{X}_i, u_i^*, d_i^*)$ to dominate the value $\mathcal{V}_i^*(\mathcal{X}_i)$ through a constant η^* . This is standard in existing literature for asymptotic tracking [53] and is also related to the controllability condition [52]. Intuitively, it ensures that the instantaneous penalty along the saddle-point solution is sufficiently strong relative to the discounted accumulation encoded in $\mathcal{V}_i^*$, which is essential for the discussion of Lyapunov stability. In practice, over $\mathbb{S}_i$, $\exists \underline{\rho}_i, \bar{v}_i \in \mathbb{R}^+$ such that $\rho_i(\mathcal{X}_i, u_i^*, d_i^*) \geq \underline{\rho}_i \|\mathcal{X}_i\|^2$ and $\mathcal{V}_i^*(\mathcal{X}_i) \leq \bar{v}_i \|\mathcal{X}_i\|^2$, then Assumption 2 holds for any $\eta^* \in (\eta, \underline{\rho}_i / \bar{v}_i]$.

Theorem 1. Let Assumptions 1 to 2 hold. Under the minimizing player $u_i^*(\mathcal{X}_i)$ in (35), the following two statements hold:

- For $d_i(t) \equiv 0$, each tracking error dynamics (24) is asymptotically stabilized.
- For the worst disturbance input $d_i^*(\mathcal{X}_i)$ in (36), the $\mathcal{L}_2$ -gain of each subsystem (26) is lesser than or equal to γ_i .

Proof. We prove this theorem by 4 steps: first, finding a proper Lyapunov candidate; second, we differentiating Lyapunov function along the closed-loop dynamics; third, applying the HJIE and Assumption 2 to obtain a strict dissipation bound, yielding asymptotic stability when $d_i \equiv 0$; finally, integrate the dissipation inequality to derive the finite-gain $\mathcal{L}_2$ stability for $d_i \neq 0$.

Firstly, from the discussions set forth, the value function $\mathcal{V}_i^*(\mathcal{X}_i)$ can be selected as the Lyapunov candidate of each augmented subsystem (26), guaranteed by the ZSO condition [32, 42]. Then, differentiating $\mathcal{V}_i^*(\mathcal{X}_i)$ along the dynamics (26) gives

$$\dot{\mathcal{V}}_i^*(\mathcal{X}_i) = \nabla^\top \mathcal{V}_i^*(\mathcal{X}_i) [\mathcal{F}_i + \mathcal{G}_i u_i^*(\mathcal{X}_i) + \mathcal{K}_i d_i^*(\mathcal{X}_i) + \Delta_i]. \quad (39)$$

From the HJIE (37), one has

$$\begin{aligned} \nabla^\top \mathcal{V}_i^*(\mathcal{X}_i) [\mathcal{F}_i + \mathcal{G}_i u_i^*(\mathcal{X}_i) + \mathcal{K}_i d_i^*(\mathcal{X}_i) + \Delta_i] \\ = -\rho_i(\mathcal{X}_i, u_i^*(\mathcal{X}_i), d_i^*(\mathcal{X}_i)) + \eta \cdot \mathcal{V}_i^*(\mathcal{X}_i). \end{aligned}$$

Then, (39) can be further expressed as

$$\begin{aligned} \dot{\mathcal{V}}_i^*(\mathcal{X}_i) &= -[\omega_i^2(\mathcal{X}_i, u_i^*) - \gamma_i^2 \cdot (d_i^*)^2] + \eta \mathcal{V}_i^*(\mathcal{X}_i) \\ &= -[q_i z_i^2 + r_i \cdot (u_i^*)^2 - \gamma_i^2 \cdot (d_i^*)^2] + \eta \mathcal{V}_i^*(\mathcal{X}_i). \end{aligned} \quad (40)$$

Based on (40), the two cases: (i) $d_i(t) \equiv 0$; (ii) $d_i^*(\mathcal{X}_i) = \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla \mathcal{V}_i^*(\mathcal{X}_i) \in \mathcal{L}_2$ are discussed below.

- i. When $d_i(t) \equiv 0$, one has $\rho_i(\mathcal{X}_i, u_i^*(\mathcal{X}_i), d_i^*(\mathcal{X}_i)) = q_i z_i^2 + r_i (u_i^*)^2$, and Assumption 2 implies

$$\mathcal{V}_i^*(\mathcal{X}_i) \leq \frac{1}{\eta^*} [q_i z_i^2 + r_i \cdot (u_i^*)^2]. \quad (41)$$

Inserting (40) and (41) into (39) gives

$$\dot{\mathcal{V}}_i^* \leq -\left(1 - \frac{\eta}{\eta^*}\right) \cdot [q_i z_i^2 + r_i \cdot (u_i^*)^2]. \quad (42)$$

With $r_i \cdot (u_i^*)^2 \geq 0$ and $\eta^* > \eta$ by Assumption 2, $\dot{\mathcal{V}}_i^*$ further satisfies

$$\dot{\mathcal{V}}_i^* \leq -q_i \left(1 - \frac{\eta}{\eta^*}\right) \cdot z_i^2. \quad (43)$$

Recall (38). Then, the derivative of $\mathcal{V}_\Sigma^*$ can be expressed by

$$\dot{\mathcal{V}}_\Sigma^* = \sum_{i=1}^n \dot{\mathcal{V}}_i^* \leq -\left(1 - \frac{\eta}{\eta^*}\right) \sum_{i=1}^n q_i z_i^2, \quad (44)$$

which indicates the asymptotic stability of the tracking error of each subsystem [2], that is, Problem 1 is tackled. Moreover, from (44), one can observe that smaller η leads to more negative $\dot{\mathcal{V}}_\Sigma^*$, which achieves smaller tracking error z_i .

- ii. When $d_i^*(\mathcal{X}_i) = \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla \mathcal{V}_i^*(\mathcal{X}_i) \in \mathcal{L}_2$, multiplying $e^{-\eta t}$ on both sides of (40) and then rearranging it, one has

$$\begin{aligned} e^{-\eta t} \dot{\mathcal{V}}_i^* - \eta e^{-\eta t} \mathcal{V}_i^* &= -e^{-\eta t} \cdot [q_i z_i^2 + r_i \cdot (u_i^*)^2] \\ &\quad + e^{-\eta t} \cdot \gamma_i^2 \cdot (d_i^*)^2, \end{aligned} \quad (45)$$

which can be equivalently expressed in a derivative form

$$\frac{d}{dt} (e^{-\eta t} \mathcal{V}_i^*) = -e^{-\eta t} \cdot [q_i z_i^2 + r_i \cdot (u_i^*)^2] + e^{-\eta t} \cdot \gamma_i^2 \cdot (d_i^*)^2. \quad (46)$$

Integrating from both sides of (46) on the interval $[0, T]$ for arbitrary $T > 0$, one can obtain

$$\begin{aligned} e^{-\eta T} \mathcal{V}_i^*(\mathcal{X}_i(T)) - \mathcal{V}_i^*(\mathcal{X}_i(0)) &= -\int_0^T e^{-\eta \tau} [q_i z_i^2 + r_i \cdot (u_i^*)^2] d\tau \\ &\quad + \gamma_i^2 \cdot \int_0^T e^{-\eta \tau} (d_i^*)^2 d\tau. \end{aligned} \quad (47)$$

Due to the positive definiteness of $\mathcal{V}_i^*$, one has $\mathcal{V}_i^*(\mathcal{X}_i(T)) \geq 0$ and $\mathcal{V}_i^*(\mathcal{X}_i(0)) \geq 0$. Then, (47) implies

$$\begin{aligned} \int_0^T e^{-\eta \tau} [q_i z_i^2 + r_i \cdot (u_i^*)^2] d\tau \\ \leq \gamma_i^2 \cdot \int_0^T e^{-\eta \tau} (d_i^*)^2 d\tau + \mathcal{V}_i^*(\mathcal{X}_i(0)). \end{aligned} \quad (48)$$

Since (48) is satisfied by each subsystem, for the whole system (24), one has

$$\begin{aligned} \int_0^T e^{-\eta \tau} \omega^2(\bar{\mathcal{X}}, \bar{u}^*) d\tau &\leq \bar{\gamma}^2 \int_0^T e^{-\eta \tau} \|\bar{d}^*\|^2 d\tau + \sum_{i=1}^n \mathcal{V}_i^*(\mathcal{X}_i(0)) \\ \Rightarrow \sqrt{\int_0^T e^{-\eta \tau} \omega^2(\bar{\mathcal{X}}, \bar{u}^*) d\tau} &\leq \bar{\gamma} \sqrt{\int_0^T e^{-\eta \tau} \|\bar{d}^*\|^2 d\tau} \\ &\quad + \sqrt{\sum_{i=1}^n \mathcal{V}_i^*(\mathcal{X}_i(0))}, \end{aligned} \quad (49)$$

where $\bar{u}^* \triangleq [u_1^*, \dots, u_n^*]$ and $\bar{d}^* \triangleq [d_1^*, \dots, d_n^*]$. From Definitions 3 and 4, it should be noted that the inequality (49) is precisely the dissipation inequality in the sense of discounted finite-gain $\mathcal{L}_2$ stability. That is, it bounds the discounted energy of the performance output by that of the disturbance plus a storage term. Moreover, under the zero-initial condition $\mathcal{X}_i(0) = 0$, (49) immediately yields the generalized discounted $\mathcal{L}_2$ -gain condition in (31), which implies the finite-gain $\mathcal{L}_2$ stability of each subsystem (26).

Collecting above discussions, one can conclude that Problems 1 and 2 are solved, and the proof is completed. $\square$

Note that the theoretical analysis in Theorem 1 is conducted on the basis that the value function $\mathcal{V}_i^*(\mathcal{X}_i)$ is known. Therefore, the H_∞ tracking control problem can be extended to the solvability of the HJIE (37) [17].

Problem 3. For each augmented subsystem (26), find the solution $\mathcal{V}_i^*(\mathcal{X}_i)$ to the corresponding HJIE (37), and thus find the Nash equilibrium $\{u_i^*, d_i^*\}$ in the form of (35) and (36).

4.2 | Actor-Critic-Disturbance Representation

To begin with, decomposing the value function $\mathcal{V}_i^*(\mathcal{X}_i)$ into two parts (known and unknown) yields:

$$\mathcal{V}_i^*(\mathcal{X}_i) = S_i(\mathcal{X}_i) + \mathfrak{F}_i^*(\mathcal{X}_i), \quad (50)$$

where $S_i(\mathcal{X}_i) \triangleq \frac{1}{2} \mathcal{X}_i^\top \Lambda_i \mathcal{X}_i$ with $\Lambda_i \triangleq \text{diag}(\kappa_i, 0)$ and user-defined parameter $\kappa_i \in \mathbb{R}^+$, and $\mathfrak{F}_i^*(\mathcal{X}_i) \triangleq \mathcal{V}_i^*(\mathcal{X}_i) - S_i(\mathcal{X}_i)$. Thus, one

has $\nabla \mathcal{V}_i^*(\mathcal{X}_i) = \nabla S_i(\mathcal{X}_i) + \nabla \mathfrak{F}_i^*(\mathcal{X}_i)$. Then, with the decomposed $\mathcal{V}_i^*(\mathcal{X}_i)$ in (50), the Nash equilibrium $\{u_i^*, d_i^*\}$ is given by

$$u_i^*(\mathcal{X}_i) = -\frac{1}{2r_i} \mathcal{G}_i^\top \nabla S_i(\mathcal{X}_i) - \frac{1}{2r_i} \mathcal{G}_i^\top \nabla \mathfrak{F}_i^*(\mathcal{X}_i), \quad (51)$$

$$d_i^*(\mathcal{X}_i) = \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla S_i(\mathcal{X}_i) + \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla \mathfrak{F}_i^*(\mathcal{X}_i). \quad (52)$$

Remark 6. Note that the HJIe (37) constitutes a nonlinear partial differential equation (PDE) with respect to the value function $\mathcal{V}_i^*(\mathcal{X}_i)$, which is inherently challenging to solve analytically. Thus, to address this difficulty, instead of approximating the entire value function $\mathcal{V}_i^*(\mathcal{X}_i)$, we concentrate on approximating the nonlinear component $\mathfrak{F}_i^*(\mathcal{X}_i)$ by the RBFNN (see Section 2.3). This strategy avoids the direct solution to the HJI PDE (37), which substantially reduces computational burden and improves practical implementability.

Remark 7. The quadratic term $S_i(\mathcal{X}_i) \triangleq \frac{1}{2} \mathcal{X}_i^\top \Lambda_i \mathcal{X}_i$ can be viewed as a locally valid ‘energy/cost’ shaping component that dominates near the origin (small tracking errors) and enforces a desired local penalty profile via the user-selected κ_i . This makes the Lyapunov analysis tractable: the known quadratic part contributes explicit negative-definite terms in the closed-loop derivative, which will be specified in the subsequent discussions.

Through the RBFNNs on the compact set $\mathbb{S}_i$, the unknown nonlinear components in (50), (51), and (52), can be expressed by neural networks (NNs), respectively, as

$$\mathfrak{F}_i^*(\mathcal{X}_i) = (\theta_i^{C*})^\top \psi_i^C(\mathcal{X}_i) + \epsilon_i^C(\mathcal{X}_i), \quad (53)$$

$$u_i^*(\mathcal{X}_i) = -\frac{1}{2r_i} \mathcal{G}_i^\top \nabla S_i(\mathcal{X}_i) + (\theta_i^{A*})^\top \psi_i^A(\mathcal{X}_i) + \epsilon_i^A(\mathcal{X}_i), \quad (54)$$

$$d_i^*(\mathcal{X}_i) = \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla S_i(\mathcal{X}_i) + (\theta_i^{D*})^\top \psi_i^D(\mathcal{X}_i) + \epsilon_i^D(\mathcal{X}_i), \quad (55)$$

where $\theta_i^{p*} \in \mathbb{R}^{N_i^p}$, $\psi_i^p : \mathbb{R}^2 \mapsto \mathbb{R}^{N_i^p}$, $\epsilon_i^p : \mathbb{R}^2 \mapsto \mathbb{R}$, and N_i^p represent the ideal weight vectors, the NN activation function vectors, the residual errors, and the number of hidden layer neurons, for $p \in \{A, C, D\}$, respectively. The function parameterization satisfies the standard assumption below.

Assumption 3. [45] The basis functions and residual errors are bounded by positive constants $\bar{\psi}_p$, $\bar{\psi}_p'$, $\bar{\epsilon}_p$, $\bar{\epsilon}_p'$, such that

$$\begin{aligned} \|\psi_i^p(\cdot)\| &\leq \bar{\psi}_p, \quad \|\nabla \psi_i^p(\cdot)\| \leq \bar{\psi}_p', \\ \|\epsilon_i^p(\cdot)\| &\leq \bar{\epsilon}_p, \quad \|\nabla \epsilon_i^p(\cdot)\| \leq \bar{\epsilon}_p', \quad \text{for each } i. \end{aligned} \quad (56)$$

Since the ideal weight vectors θ_i^{p*} remain unknown, we define the adaptive weights $\hat{\theta}_i^p \in \mathbb{R}^{N_i^p}$ to approximate θ_i^{p*} with the adaptive weight errors given by $\bar{\theta}_i^p \triangleq \theta_i^{p*} - \hat{\theta}_i^p$. Accordingly, on the compact set $\mathbb{S}_i$, (53), (54), and (55) can be approximated by the adaptive Critic, Actor, and Disturbance NNs, respectively, as

$$\hat{\mathfrak{F}}_i(\mathcal{X}_i) = (\hat{\theta}_i^C)^\top \psi_i^C(\mathcal{X}_i), \quad (57)$$

$$\hat{u}_i^A(\mathcal{X}_i) = -\frac{1}{2r_i} \mathcal{G}_i^\top \nabla S_i(\mathcal{X}_i) + (\hat{\theta}_i^A)^\top \psi_i^A(\mathcal{X}_i), \quad (58)$$

$$\hat{d}_i^D(\mathcal{X}_i) = \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla S_i(\mathcal{X}_i) + (\hat{\theta}_i^D)^\top \psi_i^D(\mathcal{X}_i). \quad (59)$$

It should be noted that $\hat{u}_i^A$ for $i = 1, 2, \dots, n-1$ are virtual controllers for intermediate steps. The real control (i.e., the actual Actor) of the system (17) can be expressed explicitly as

$$\begin{aligned} u &= \hat{u}_n^A = -\frac{1}{2r_n} \mathcal{G}_n^\top \nabla S_n(\mathcal{X}_n) + (\hat{\theta}_n^A)^\top \psi_n^A(\mathcal{X}_n) \\ &= -\frac{\kappa_n}{2r_n} g_n z_n + (\hat{\theta}_n^A)^\top \psi_n^A(\mathcal{X}_n). \end{aligned}$$

Building upon the preceding discussion, the subsequent subsections are devoted to formulating the Actor-Critic-Disturbance learning algorithm, accompanied by rigorous theoretical analysis on both the stability and convergence of the tracking error and adaptive weights.

4.3 | Robust Adaptive Actor-Critic-Disturbance Synchronous Online Learning

In this subsection, we develop the adaptive Actor-Critic-Disturbance synchronous online learning structure. As illustrated in Figure 1, the n -dimensional nonlinear strict-feedback system is decoupled into n augmented subsystems (see Section 3). Each subsystem evolves within the Actor-Critic-Disturbance learning structure through the experience replay mechanism, which is closely related to the following definition.

Definition 6. [34, 35] A bounded vector signal $\rho(t) \in \mathbb{R}^m$ is said to be:

1. PE, if $\exists \bar{\rho}, \mathcal{T} \in \mathbb{R}^+$ such that $\int_t^{t+\mathcal{T}} \rho(t) \rho^\top(t) dt \geq \bar{\rho} \cdot \mathbf{I}_m, \forall t \geq 0$;
2. SE over an interval, if $\exists \bar{\rho}, \mathcal{T} \in \mathbb{R}^+$, and a time sequence $\{t_j\}_{j=1}^L$ with positive integer L , such that $\sum_{j=1}^L \rho(t_j) \rho^\top(t_j) \geq \bar{\rho} \cdot \mathbf{I}_m$ for $t_j \in (0, \mathcal{T})$.

It should pointed out that the SE condition is a relaxation of PE, which is required in conventional ADP design to guarantee the convergence of the adaptive weight. But this can be difficult in practical implementation, as verifying PE requires continuous excitation over infinite time horizon and may not always be feasible particularly in applications that require high precision or smooth operation [35]. The SE condition is instead built on the reuse of sufficiently rich data collected at past time instants, which can still achieve convergence and thus weaken the requirement of PE [37]. Therefore, in this article, we enforce the SE condition that employs historical data, concurrently with the online data, as detailed subsequently.

4.3.1 | Critic Learning

In the following discussion, for brevity, we omit the arguments of functions whenever no ambiguity arises, for example, $\hat{u}_i = \hat{u}_i(\mathcal{X}_i)$.

Under the input players $\hat{u}_i$ and $\hat{d}_i$ given by (58) and (59), respectively, the Hamiltonian $\mathcal{H}_i$ in (37) is given by

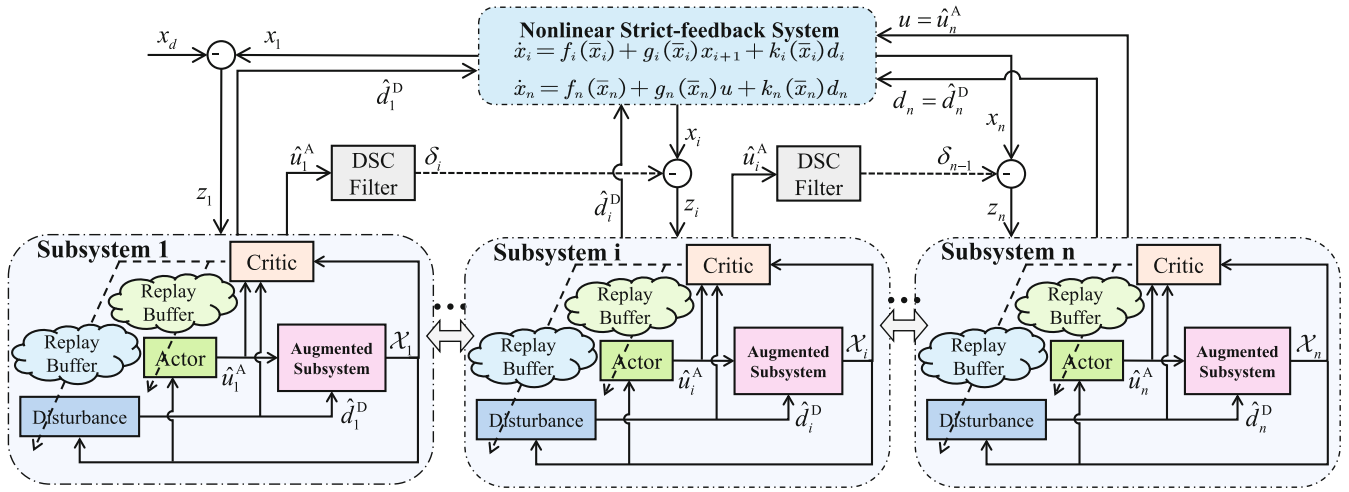


FIGURE 1 | The block diagram of the presented Actor-Critic-Disturbance learning algorithm.

$$\mathcal{H}_i(\mathcal{X}_i, \hat{u}_i^A, \hat{d}_i^D, \nabla \hat{V}_i) = \rho_i(\mathcal{X}_i, \hat{u}_i^A, \hat{d}_i^D) + \nabla^\top \hat{V}_i (F_i + G_i \hat{u}_i^A + K_i \hat{d}_i^D + \Delta_i) - \eta \hat{V}_i, \quad (60)$$

where $\hat{V}_i \triangleq S_i + \hat{\mathfrak{S}}_i$. Inserting (57) into (60) yields

$$\mathcal{H}_i(\mathcal{X}_i, \hat{u}_i^A, \hat{d}_i^D, \nabla \hat{V}_i) = (\hat{\theta}_i^C)^\top \xi_i^C + \rho_i(\mathcal{X}_i, \hat{u}_i^A, \hat{d}_i^D) - \eta S_i + \nabla^\top S_i (F_i + G_i \hat{u}_i^A + K_i \hat{d}_i^D + \Delta_i), \quad (61)$$

where $\xi_i^C \triangleq \nabla \psi_i^C(F_i + G_i \hat{u}_i^A + K_i \hat{d}_i^D) - \eta \psi_i^C$. Accordingly, the Critic learning error at time t for each subsystem is then defined as

$$e_i^C(t) \triangleq \mathcal{H}_i(\mathcal{X}_i, \hat{u}_i^A, \hat{d}_i^D, \nabla \hat{V}_i) - \mathcal{H}_i(\mathcal{X}_i, u_i^*, d_i^*, \nabla V_i^*) = -(\hat{\theta}_i^C)^\top(t) \cdot \xi_i^C(t) + \beta_i^C(t), \quad (62)$$

where

$$\beta_i^C(t) \triangleq (\theta_i^{C*})^\top \cdot \xi_i^C + \rho_i(\mathcal{X}_i, \hat{u}_i^A(\mathcal{X}_i), \hat{d}_i^D(\mathcal{X}_i)) - \eta S_i(\mathcal{X}_i) + \nabla^\top S_i(\mathcal{X}_i) [F_i + G_i \hat{u}_i^A(\mathcal{X}_i) + K_i \hat{d}_i^D(\mathcal{X}_i) + \Delta_i]. \quad (63)$$

As pointed out by existing literature [6, 54, 55], as the number of Critic neurons $N_i^C \rightarrow \infty$, one has β_i^C satisfy $\sup |\beta_i^C| \rightarrow 0$, which implies the larger N_i^C , the smaller the upper bound of β_i^C . Moreover, the Critic learning aims to achieve: $\hat{\theta}_i^C(t) \rightarrow \theta_i^{C*}$ as $t \rightarrow \infty$, then $\tilde{\theta}_i^C(t) \rightarrow 0$, and the Critic error $e_i^C(t) \rightarrow \beta_i^C(t)$. This implies the Critic error $e_i^C(t)$ can be made arbitrarily small by accurate learning and sufficiently large N_i^C .

Then, to weaken the PE condition, at past instants $t_j^C < t$ for $j = 1, \dots, M_i^C$, define the corresponding composite Critic error

$$e_i^{C_j}(t) = -(\tilde{\theta}_i^C)^\top(t) \cdot \xi_i^{C_j} + \beta_i^{C_j}, \quad (64)$$

where $\xi_i^{C_j} \triangleq \xi_i^C(t_j^C)$ and $\beta_i^{C_j} \triangleq \beta_i^C(t_j^C)$.

Then, based on both the ongoing and past Critic errors (62) and (64), respectively, define the objective function to be minimized by each Critic as

$$\mathcal{E}_i^C \triangleq \frac{1}{2} \left[\frac{(e_i^C)^2}{(1 + \|\xi_i^C\|^2)^2} + \sum_{j=1}^{M_i^C} \frac{(e_i^{C_j})^2}{(1 + \|\xi_i^{C_j}\|^2)^2} \right]. \quad (65)$$

For each subsystem, we maintain a replay buffer $\mathcal{B}_i^C$ with capacity M_i^C that stores tuples $\{\mathcal{X}_i, \xi_i^C\}$ (and any other required intermediate terms). At each time step, the newest tuple is appended; if the buffer is full, the oldest entry is discarded. During each update, we sample M_i^C past entries uniformly at random from the buffer $\mathcal{B}_i^C$ to form the replay summation in (65).

From (65), the Critic learning law is proposed using gradient descent as

$$\begin{aligned} \hat{\theta}_i^C &= -\alpha_i^C \cdot \frac{\partial \mathcal{E}_i^C}{\partial \hat{\theta}_i^C} \\ &= -\alpha_i^C \cdot \left[\frac{\xi_i^C e_i^C}{(1 + \|\xi_i^C\|^2)^2} + \sum_{j=1}^{M_i^C} \frac{\xi_i^{C_j} e_i^{C_j}}{(1 + \|\xi_i^{C_j}\|^2)^2} \right], \end{aligned} \quad (66)$$

where $\alpha_i^C \in \mathbb{R}^+$ is the Critic learning rate for each i .

4.3.2 | Actor Learning

Under the Critic NNs (57), define the guidance for the Actor as

$$\begin{aligned} \hat{u}_i^C(\mathcal{X}_i) &\triangleq -\frac{1}{2r_i} \mathcal{G}_i^\top \nabla S_i(\mathcal{X}_i) - \frac{1}{2r_i} \mathcal{G}_i^\top \nabla \hat{\mathfrak{S}}_i(\mathcal{X}_i) \\ &= -\frac{1}{2r_i} \mathcal{G}_i^\top \nabla S_i(\mathcal{X}_i) - \frac{1}{2r_i} \mathcal{G}_i^\top \nabla^\top \psi_i^C(\mathcal{X}_i) \hat{\theta}_i^C, \end{aligned} \quad (67)$$

by which we can define the Actor learning error as

$$\begin{aligned} \mathbf{e}_i^A(t) \triangleq \hat{\mathbf{u}}_i^A(\mathcal{X}_i) - \hat{\mathbf{u}}_i^C(\mathcal{X}_i) = & -(\tilde{\theta}_i^A)^\top(t) \cdot \psi_i^A(\mathcal{X}_i(t)) \\ & + \underbrace{(\theta_i^{A*})^\top \cdot \psi_i^A(\mathcal{X}_i(t)) + \frac{1}{2r_i} \mathcal{G}_i^\top \nabla^\top \psi_i^C(\mathcal{X}_i(t)) \hat{\theta}_i^C(t)}_{\triangleq \beta_i^A(t)}. \end{aligned} \quad (68)$$

Similar to the Critic case in (64), at time points $t_j^A < t$ for $j = 1, \dots, M_i^A$, the composite Actor learning error is then given by

$$\mathbf{e}_i^{A_j}(t) \triangleq -(\tilde{\theta}_i^A)^\top(t) \cdot \psi_i^{A_j} + \beta_i^{A_j}. \quad (69)$$

where $\psi_i^{A_j} \triangleq \psi_i^A(\mathcal{X}_i(t_j^A))$ and $\beta_i^{A_j} \triangleq \beta_i^A(t_j)$ for simplicity. Similar to the Critic learning, we sample M_i^A past entries uniformly at random from a buffer $\mathcal{B}_i^A$ to form the replay summation. Then, each Actor aims to minimize the objective function

$$\mathcal{E}_i^A \triangleq \frac{1}{2} \left[(\mathbf{e}_i^A)^2 + \sum_{j=1}^{M_i^A} (\mathbf{e}_i^{A_j})^2 \right], \quad (70)$$

from which the Actor update law is developed as

$$\dot{\theta}_i^A = -\alpha_i^A \cdot \frac{\partial \mathcal{E}_i^A}{\partial \hat{\theta}_i^A} = -\alpha_i^A \cdot \left(\psi_i^A \mathbf{e}_i^A + \sum_{j=1}^{M_i^A} \psi_i^{A_j} \mathbf{e}_i^{A_j} \right) \quad \text{for each } i \quad (71)$$

with the Actor learning rate $\alpha_i^A \in \mathbb{R}^+$.

4.3.3 | Disturbance Learning

The target of the Disturbance is defined via the Critic NNs as

$$\hat{d}_i^C(\mathcal{X}_i) \triangleq \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla S_i(\mathcal{X}_i) + \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla^\top \psi_i^C(\mathcal{X}_i) \hat{\theta}_i^C. \quad (72)$$

Accordingly, the Disturbance learning error is formed as

$$\begin{aligned} \mathbf{e}_i^D(t) \triangleq \hat{d}_i^D(\mathcal{X}_i) - \hat{d}_i^C(\mathcal{X}_i) = & -(\tilde{\theta}_i^D)^\top(t) \cdot \psi_i^D(\mathcal{X}_i(t)) \\ & + \underbrace{(\theta_i^{D*})^\top(t) \cdot \psi_i^D(\mathcal{X}_i(t)) - \frac{1}{2\gamma_i^2} \mathcal{K}_i^\top \nabla^\top \psi_i^C(\mathcal{X}_i(t)) \hat{\theta}_i^C(t)}_{\triangleq \beta_i^D(t)}. \end{aligned} \quad (73)$$

At historical sample instants $t_j^D < t$ for $j = 1, \dots, M_i^D$, the composite Disturbance learning error is

$$\mathbf{e}_i^{D_j}(t) \triangleq -(\tilde{\theta}_i^D)^\top(t) \cdot \psi_i^{D_j} + \beta_i^{D_j} \quad (74)$$

where $\psi_i^{D_j} \triangleq \psi_i^D(\mathcal{X}_i(t_j^D))$ and $\beta_i^{D_j} \triangleq \beta_i^D(t_j)$ for brevity. To relax the PE requirement, we sample M_i^D past entries uniformly at random from a buffer $\mathcal{B}_i^D$ to form the replay summation. Then, the Disturbance objective function is given by

$$\mathcal{E}_i^D \triangleq \frac{1}{2} \left[(\mathbf{e}_i^D)^2 + \sum_{j=1}^{M_i^D} (\mathbf{e}_i^{D_j})^2 \right]. \quad (75)$$

ALGORITHM 1 | Robust Actor-Critic-Disturbance Learning for H_∞ Tracking.

- 1: **Input:** user-defined parameters $\{q_i, r_i, \gamma_i, \kappa_i, \tau_i, \eta\}$, learning rates $\{\alpha_i^C, \alpha_i^A, \alpha_i^D\}$, basis functions $\{\psi_i^C, \psi_i^A, \psi_i^D\}$, buffer size $\{M_i^C, M_i^A, M_i^D\}$, reference signal x_d .
- 2: **Initialization:** original system states $x_i(0)$, network weights $\{\theta_i^C(0), \theta_i^A(0), \theta_i^D(0)\}$, DSC internal states $\delta_i(0)$, replay buffers $\{\mathcal{B}_i^C, \mathcal{B}_i^A, \mathcal{B}_i^D\}$, running time $[0, t_{end}]$.
- 3: **for** $t \in [0, t_{end}]$ **do**
- 4: **for** each i from 1 to N **do**
- 5: Measure state $x_i(t)$; compute tracking error $z_i(t)$ and augmented state $\mathcal{X}_i(t)$.
- 6: Compute the input players $\{\hat{u}_i^A, \hat{d}_i^D\}$ by (58) and (59) for the ZSG (33).
- 7: Compute the Critic regressor ξ_i^C in (61).
- 8: **for** $p \in \{A, C, D\}$ **do**
- 9: Update replay buffer: $\mathcal{B}_i^p$.
- 10: **end for**
- 11: Update the Critic weight by (66).
- 12: Update the Actor and Disturbance weights by (71) and (76).
- 13: **end for**
- 14: Propagate t and $x_i(t)$ for $i = 1, 2, \dots, n$ along (17) with $u = \hat{u}_n^A(\mathcal{X}_n)$ and $d_n = \hat{d}_n^D(\mathcal{X}_n)$.
- 15: **end for**

By gradient descent, the Disturbance tuning law is established as

$$\dot{\theta}_i^D = -\alpha_i^D \cdot \frac{\partial \mathcal{E}_i^D}{\partial \hat{\theta}_i^D} = -\alpha_i^D \cdot \left(\psi_i^D \mathbf{e}_i^D + \sum_{j=1}^{M_i^D} \psi_i^{D_j} \mathbf{e}_i^{D_j} \right) \quad \text{for each } i \quad (76)$$

with the learning rate $\alpha_i^D \in \mathbb{R}^+$. To facilitate implementation and provide a clear overview of the online learning workflow presented in Section 4.3, we summarize the robust adaptive Actor-Critic-Disturbance learning procedure in Algorithm 1. Specifically, the algorithm organizes the real-time data acquisition, replay-buffer updates, and weight adaptation laws into a unified framework.

The stability and convergence of the proposed learning scheme are established in the following subsection.

4.4 | Stability and Convergence Analysis

Hereafter, under the Actor-Critic-Disturbance learning scheme in Section 4.3, the overall stability analysis of the closed-loop system follows sequentially, beginning with the adaptive weight errors and subsequently addressing the tracking errors.

4.4.1 | Critic Weight Error Dynamics

Based on (71), the dynamics of Critic weight error is given by

$$\dot{\hat{\theta}}_i^C = -\hat{\theta}_i^C = \alpha_i^C \cdot \left[\frac{\xi_i^C \mathbf{e}_i^C}{(1 + \|\xi_i^C\|^2)^2} + \sum_{j=1}^{M_i^C} \frac{\xi_i^{C_j} \mathbf{e}_i^{C_j}}{(1 + \|\xi_i^{C_j}\|^2)^2} \right], \quad (77)$$

from which one can obtain the derivative of the Lyapunov candidate $S_i^C(\tilde{\theta}_i^C) \triangleq \frac{1}{2\alpha_i^C} \|\tilde{\theta}_i^C\|^2$ as

$$\dot{S}_i^C(\tilde{\theta}_i^C) = (\tilde{\theta}_i^C)^\top \left[\frac{\xi_i^C e_i^C}{(1 + \|\xi_i^C\|^2)^2} + \sum_{j=1}^{M_i^C} \frac{\xi_i^{C_j} e_i^{C_j}}{(1 + \|\xi_i^{C_j}\|^2)^2} \right]. \quad (78)$$

Inserting (62) and (64) into (78) yields

$$\dot{S}_i^C = -(\tilde{\theta}_i^C)^\top (\Phi_1 + \Phi_2) \tilde{\theta}_i^C + (\tilde{\theta}_i^C)^\top \left(\Phi_{i,3}^C \beta_i^C + \sum_{j=1}^{M_i^C} \Phi_{i,4}^{C_j} \beta_i^{C_j} \right), \quad (79)$$

where

$$\begin{aligned} \Phi_{i,1}^C &\triangleq \frac{\xi_i^C (\xi_i^C)^\top}{(1 + \|\xi_i^C\|^2)^2}, \Phi_{i,2}^C \triangleq \sum_{j=1}^{M_i^C} \frac{\xi_i^{C_j} (\xi_i^{C_j})^\top}{(1 + \|\xi_i^{C_j}\|^2)^2}, \\ \Phi_{i,3}^C &\triangleq \frac{\xi_i^C}{(1 + \|\xi_i^C\|^2)^2}, \Phi_{i,4}^{C_j} \triangleq \frac{\xi_i^{C_j}}{(1 + \|\xi_i^{C_j}\|^2)^2}. \end{aligned} \quad (80)$$

Fact 1. For any arbitrary vector ζ , the following inequalities hold:

$$\left\| \frac{1}{1 + \|\zeta\|^2} \right\| \leq 1, \quad \left\| \frac{\zeta}{1 + \|\zeta\|^2} \right\| \leq \frac{1}{2}. \quad (81)$$

Then, the stability of each Critic weight error can be summarized as follows.

Theorem 2. Let Assumptions 1 to 3, hold. Under the input tuple $\{\hat{u}_i^A, \hat{d}_i^D\}$ given by (58) and (59) with the Actor and Disturbance learning laws presented in (71) and (76), respectively, consider the Critic tuning law presented in (66) with the residual error β_i^C being bounded by $\bar{\beta}_i^C \in \mathbb{R}^+$, such that $|\beta_i^C| \leq \bar{\beta}_i^C$. Then, the adaptive weight error $\tilde{\theta}_i^C$ of each Critic NN is uniformly ultimately bounded (UUB), provided that the signals $\xi_i^C(t)$ satisfy the SE condition.

Proof. First, combined Fact 1 with $|\beta_i^C| \leq \bar{\beta}_i^C$, the term $(\tilde{\theta}_i^C)^\top (\Phi_{i,3}^C \beta_i^C + \sum_{j=1}^{M_i^C} \Phi_{i,4}^{C_j} \beta_i^{C_j})$ in (79) satisfies

$$(\tilde{\theta}_i^C)^\top \left(\Phi_{i,3}^C \beta_i^C + \sum_{j=1}^{M_i^C} \Phi_{i,4}^{C_j} \beta_i^{C_j} \right) \leq \frac{(1 + M_i^C) \bar{\beta}_i^C}{2} \cdot \|\tilde{\theta}_i^C\|. \quad (82)$$

Since the signal $\xi_i^C(t)$ is SE, that is, the matrix $\sum_{j=1}^{M_i^C} \xi_i^{C_j} (\xi_i^{C_j})^\top$ is full rank and positive definite, then, with $\frac{1}{(1 + \|\xi_i^{C_j}\|^2)^2} \leq 1$ (see

Fact 1), there exist some $\lambda_i^C \in \mathbb{R}^+$, such that

$$\Phi_{i,2}^C = \sum_{j=1}^{M_i^C} \frac{\xi_i^{C_j} (\xi_i^{C_j})^\top}{(1 + \|\xi_i^{C_j}\|^2)^2} \geq \lambda_i^C \cdot \mathbf{I}_{N_i^C} \Leftrightarrow \lambda_{\min}(\Phi_{i,2}^C) \geq \lambda_i^C. \quad (83)$$

Moreover, $\Phi_{i,1}^C$ is positive semi-definite, which implies $\lambda_{\min}(\Phi_{i,1}^C + \Phi_{i,2}^C) \geq \lambda_i^C$. Then, with (82) and (83), $\dot{S}_i^C$ in (79) further satisfies

$$\begin{aligned} \dot{S}_i^C &\leq -\lambda_i^C \|\tilde{\theta}_i^C\|^2 + \frac{(1 + M_i^C) \bar{\beta}_i^C}{2} \cdot \|\tilde{\theta}_i^C\| \\ &= -b_i^C \lambda_i^C \|\tilde{\theta}_i^C\|^2 - (1 - b_i^C) \lambda_i^C \|\tilde{\theta}_i^C\|^2 \\ &\quad + \frac{(1 + M_i^C) \bar{\beta}_i^C}{2} \cdot \|\tilde{\theta}_i^C\| \quad \text{for } b_i^C \in (0, 1). \end{aligned} \quad (84)$$

In addition, as the node of the neurons N_i^C increases, the bound $\bar{\beta}_i^C$ contained in (84) decreases [45, 55]. Hence, by the UUB theorem [2], after some finite time T_i^C , the Critic adaptive weight error converges to a small neighborhood of the origin, such that $\tilde{\theta}_i^C(t) \in \Omega_i^C$ for $t \geq T_i^C$, where

$$\begin{aligned} \Omega_i^C &\triangleq \left\{ \tilde{\theta}_i^C(t) \in \mathbb{R}^{N_i^C} \mid \|\tilde{\theta}_i^C(t)\| \leq \Theta_i^C \right\}, \\ \Theta_i^C &\triangleq \frac{(1 + M_i^C) \bar{\beta}_i^C}{2(1 - b_i^C) \lambda_i^C}. \end{aligned} \quad (85)$$

This completes the proof. $\square$

4.4.2 | Actor and Disturbance Weight Error Dynamics

The subsequent discussion investigates the dynamics of Actor and Disturbance weight errors simultaneously, which can be expressed using (71) and (76), respectively, as

$$\dot{\tilde{\theta}}_i^A = -\dot{\tilde{\theta}}_i^A = \alpha_i^A \cdot \frac{\partial \mathcal{E}_i^A}{\partial \tilde{\theta}_i^A} = \alpha_i^A \cdot \left(\psi_i^A e_i^A + \sum_{j=1}^{M_i^A} \psi_i^{A_j} e_i^{A_j} \right), \quad (86)$$

$$\dot{\tilde{\theta}}_i^D = -\dot{\tilde{\theta}}_i^D = \alpha_i^D \cdot \frac{\partial \mathcal{E}_i^D}{\partial \tilde{\theta}_i^D} = \alpha_i^D \cdot \left(\psi_i^D e_i^D + \sum_{j=1}^{M_i^D} \psi_i^{D_j} e_i^{D_j} \right). \quad (87)$$

Recall the errors (68), (69), (73), and (74) in the composite learning process. With the Lyapunov candidates chosen as $S_i^A(\tilde{\theta}_i^A) \triangleq \frac{1}{2\alpha_i^A} \|\tilde{\theta}_i^A\|^2$ and $S_i^D(\tilde{\theta}_i^D) \triangleq \frac{1}{2\alpha_i^D} \|\tilde{\theta}_i^D\|^2$, one can obtain

$$\begin{aligned} \dot{S}_i^A(\tilde{\theta}_i^A) &= -(\tilde{\theta}_i^A)^\top (\Phi_{i,1}^A + \Phi_{i,2}^A) \tilde{\theta}_i^A \\ &\quad + (\tilde{\theta}_i^A)^\top \left(\psi_i^A \beta_i^A + \sum_{j=1}^{M_i^A} \psi_i^{A_j} \beta_i^{A_j} \right), \end{aligned} \quad (88)$$

$$\begin{aligned} \dot{S}_i^D(\tilde{\theta}_i^D) &= -(\tilde{\theta}_i^D)^\top (\Phi_{i,1}^D + \Phi_{i,2}^D) \tilde{\theta}_i^D \\ &\quad + (\tilde{\theta}_i^D)^\top \left(\psi_i^D \beta_i^D + \sum_{j=1}^{M_i^D} \psi_i^{D_j} \beta_i^{D_j} \right), \end{aligned} \quad (89)$$

where

$$\begin{aligned} \Phi_{i,1}^A &\triangleq \psi_i^A (\psi_i^A)^\top, \Phi_{i,2}^A \triangleq \sum_{j=1}^{M_i^A} \psi_i^{A_j} (\psi_i^{A_j})^\top, \\ \Phi_{i,1}^D &\triangleq \psi_i^D (\psi_i^D)^\top, \Phi_{i,2}^D \triangleq \sum_{j=1}^{M_i^D} \psi_i^{D_j} (\psi_i^{D_j})^\top. \end{aligned} \quad (90)$$

Theorem 3. Under the same conditions in Theorem 2, let the signals $\psi_i^A(t)$ and $\psi_i^D(t)$ be SE. Then, the following results hold:

- The adaptive weight errors of each Actor and Disturbance are UUB.
- The input pair $\{\hat{u}_i^A, \hat{d}_i^D\}$ in (71) and (76) gives the approximate Nash solution to the ZSG (33), that is, Problem 3 is solved.

Proof. To begin with, we first prove the UUB property of $\tilde{\theta}_i^A$ and $\tilde{\theta}_i^D$.

(i) Since the SE condition is satisfied by $\psi_i^A(t)$ and $\psi_i^D(t)$, as discussed in the Critic case, for example, see (83), there exist some $\lambda_i^A \in \mathbb{R}^+$ and $\lambda_i^D \in \mathbb{R}^+$, such that

$$\lambda_{\min}(\Phi_{i,2}^A) \geq \lambda_i^A, \quad \lambda_{\min}(\Phi_{i,2}^D) \geq \lambda_i^D. \quad (91)$$

With the fact that $\Phi_{i,1}^A, \Phi_{i,1}^D$ being positive semi-definite, the following holds

$$\lambda_{\min}(\Phi_{i,1}^A + \Phi_{i,2}^A) \geq \lambda_i^A, \quad \lambda_{\min}(\Phi_{i,1}^D + \Phi_{i,2}^D) \geq \lambda_i^D. \quad (92)$$

Then, what remain to discuss are the residual terms β_i^A and β_i^D defined in (68) and (73), respectively, which both contain $\hat{\theta}_i^C$. As proved in Theorem 2, one has $\|\hat{\theta}_i^C\| = \|\theta_i^{C*} - \tilde{\theta}_i^C\| \leq \|\theta_i^{C*}\| + \Theta_i^C$. Then, under Assumption 3 and (23), β_i^A and β_i^D are upper bounded satisfying

$$\begin{aligned} |\beta_i^A| &\leq \bar{\beta}_i^A \triangleq \|\theta_i^{A*}\| \bar{\psi}_A + \frac{1}{2r_i} \bar{g} \bar{\psi}'_C (\|\theta_i^{C*}\| + \Theta_i^C), \\ |\beta_i^D| &\leq \bar{\beta}_i^D \triangleq \|\theta_i^{D*}\| \bar{\psi}_D + \frac{1}{2\gamma_i^2} \bar{k} \bar{\psi}'_C (\|\theta_i^{C*}\| + \Theta_i^C). \end{aligned} \quad (93)$$

Substituting (92) and (93) into (88) and (89) yields

$$\begin{aligned} \dot{S}_i^A &\leq -\lambda_i^A \|\tilde{\theta}_i^A\|^2 + \|\tilde{\theta}_i^A\| \cdot (1 + M_i^A) \bar{\beta}_i^A \\ &= -b_i^A \lambda_i^A \|\tilde{\theta}_i^A\|^2 - (1 - b_i^A) \lambda_i^A \|\tilde{\theta}_i^A\|^2 + \|\tilde{\theta}_i^A\| \\ &\quad \cdot (1 + M_i^A) \bar{\beta}_i^A \quad \text{for } b_i^A \in (0, 1), \end{aligned} \quad (94)$$

$$\begin{aligned} \dot{S}_i^D &\leq -\lambda_i^D \|\tilde{\theta}_i^D\|^2 + \|\tilde{\theta}_i^D\| \cdot (1 + M_i^D) \bar{\beta}_i^D \\ &= -b_i^D \lambda_i^D \|\tilde{\theta}_i^D\|^2 - (1 - b_i^D) \lambda_i^D \|\tilde{\theta}_i^D\|^2 + \|\tilde{\theta}_i^D\| \\ &\quad \cdot (1 + M_i^D) \bar{\beta}_i^D \quad \text{for } b_i^D \in (0, 1). \end{aligned} \quad (95)$$

For $i = 1, \dots, n$, following the UUB theorem [2], one can conclude that there exist some finite time T_i^A for the Actor and T_i^D for the Disturbance with $\min\{T_i^A, T_i^D\} \geq T_i^C$, such that $\tilde{\theta}_i^A$ and $\tilde{\theta}_i^D$ both converge to small neighborhoods of the origin: $\tilde{\theta}_i^A(t) \in \Omega_i^A$ for $t \geq T_i^A$ and $\tilde{\theta}_i^D(t) \in \Omega_i^D$ for $t \geq T_i^D$, where

$$\Omega_i^A \triangleq \left\{ \tilde{\theta}_i^A(t) \in \mathbb{R}^{N_i^A} \mid \|\tilde{\theta}_i^A(t)\| \leq \Theta_i^A \right\}, \quad \Theta_i^A \triangleq \frac{(1 + M_i^A) \bar{\beta}_i^A}{2(1 - b_i^A) \lambda_i^A} \quad (96)$$

$$\Omega_i^D \triangleq \left\{ \tilde{\theta}_i^D(t) \in \mathbb{R}^{N_i^D} \mid \|\tilde{\theta}_i^D(t)\| \leq \Theta_i^D \right\}, \quad \Theta_i^D \triangleq \frac{(1 + M_i^D) \bar{\beta}_i^D}{2(1 - b_i^D) \lambda_i^D} \quad (97)$$

Then, the first result is obtained.

(ii) Hereafter, we demonstrate the convergence of $\{\hat{u}_i^A, \hat{d}_i^D\}$ to the Nash equilibrium $\{u_i^*, d_i^*\}$. Under Assumption 3, considering (54), (55), (58), and (59), one has

$$|\hat{u}_i^A - u_i^*| = \left| -\tilde{\theta}_i^A \psi_i^A - \epsilon_i^A \right| \leq \|\tilde{\theta}_i^A\| \cdot \bar{\psi}_A + \bar{\epsilon}_A, \quad (98)$$

$$|\hat{d}_i^D - d_i^*| = \left| -\tilde{\theta}_i^D \psi_i^D - \epsilon_i^D \right| \leq \|\tilde{\theta}_i^D\| \cdot \bar{\psi}_D + \bar{\epsilon}_D. \quad (99)$$

Since the weight errors $\tilde{\theta}_i^A$ and $\tilde{\theta}_i^D$ are UUB, one can conclude that for each i , the input pair $\{\hat{u}_i^A, \hat{d}_i^D\}$ converge to a small neighborhood of the Nash equilibrium to the ZSG (33). This completes the proof. $\square$

Remark 8. The experience replay mechanism forms the “information matrices” $\Phi_{i,2}^C, \Phi_{i,2}^A$, and $\Phi_{i,2}^D$, which collect the historical data during the online learning process. Guaranteed by the SE condition such that the information contained by $\{\Phi_{i,2}^C, \Phi_{i,2}^A, \Phi_{i,2}^D\}$ is sufficiently rich, positiveness of the minimum eigenvalues of the information matrices can be ensured, as stated in (83) and (91). This immediately leads to the Lyapunov inequalities in (84), (94), and (95), which implies the UUB property of the adaptive weight errors $\{\tilde{\theta}_i^C, \tilde{\theta}_i^A, \tilde{\theta}_i^D\}$.

4.4.3 | Tracking Error Dynamics

Considering the two players given by (58) and (59), one can express the dynamics of the tracking error (24) by

$$\begin{aligned} \dot{z}_i &= f_i + g_i \hat{u}_i^A + k_i \hat{d}_i^D - \ell_{i-1}(\delta_{i-1}) \\ &= f_i + g_i \left[-\frac{1}{2r_i} \mathcal{G}_i^T \nabla S_i + \left(\hat{\theta}_i^A \right)^T \psi_i^A \right] \\ &\quad + k_i \left[\frac{1}{2\gamma_i^2} \mathcal{K}_i^T \nabla S_i + \left(\hat{\theta}_i^D \right)^T \psi_i^D \right] - \ell_{i-1}(\delta_{i-1}). \end{aligned} \quad (100)$$

For stability analysis of the tracking error and the DSC filter error, we employ

$$S(\bar{z}, \bar{\delta}) = \sum_{i=1}^n S_i(z_i) + \sum_{i=1}^{n-1} S_i^{\delta}(\tilde{\delta}_i) \quad (101)$$

as the Lyapunov candidate, where $S_i^{\delta} \triangleq \frac{1}{2} \tilde{\delta}_i^2$, $\bar{z} \triangleq [z_1, \dots, z_n]$, and $\bar{\delta} \triangleq [\tilde{\delta}_1, \dots, \tilde{\delta}_{n-1}]$.

For $i = 1, \dots, n-1$, along the error dynamic (100), the derivative of S_i is given by

$$\begin{aligned} \dot{S}_i &= \kappa_i z_i \dot{z}_i = \kappa_i z_i \left\{ f_i + g_i \left[-\frac{1}{2r_i} \mathcal{G}_i^T \nabla S_i + \left(\hat{\theta}_i^A \right)^T \psi_i^A \right] \right. \\ &\quad \left. + k_i \left[\frac{1}{2\gamma_i^2} \mathcal{K}_i^T \nabla S_i + \left(\hat{\theta}_i^D \right)^T \psi_i^D \right] - \ell_{i-1}(\delta_{i-1}) + g_i z_{i+1} + g_i \tilde{\delta}_{i+1} \right\} \end{aligned}$$

$$\begin{aligned}
&= -\frac{\kappa_i^2}{2} \left(\frac{1}{r_i} g_i^2 - \frac{1}{2\gamma_i^2} k_i^2 \right) z_i^2 - \kappa_i z_i \left[g_i \left(\tilde{\theta}_i^A \right)^\top \psi_i^A + k_i \left(\tilde{\theta}_i^D \right)^\top \psi_i^D \right] \\
&\quad + \kappa_i z_i \left[f_i - \ell_{i-1}(\delta_{i-1}) \right] \\
&\quad + \kappa_i z_i \left[g_i \left(\tilde{\theta}_i^{A*} \right)^\top \psi_i^A + k_i \left(\tilde{\theta}_i^{D*} \right)^\top \psi_i^D \right] + \kappa_i g_i z_i z_{i+1} + \kappa_i g_i z_i \tilde{\delta}_i.
\end{aligned} \tag{102}$$

For z_n , similarly, one has

$$\begin{aligned}
\dot{S}_n &= -\frac{\kappa_n^2}{2} \left(\frac{1}{r_n} g_n^2 - \frac{1}{2\gamma_n^2} k_n^2 \right) z_n^2 - \kappa_n z_n \left[g_n \left(\tilde{\theta}_n^A \right)^\top \psi_n^A + k_n \left(\tilde{\theta}_n^D \right)^\top \psi_n^D \right] \\
&\quad + \kappa_n z_n \left[f_n - \ell_{n-1}(\delta_{n-1}) \right] + \kappa_n z_n \left[g_n \left(\tilde{\theta}_n^{A*} \right)^\top \psi_n^A + k_n \left(\tilde{\theta}_n^{D*} \right)^\top \psi_n^D \right].
\end{aligned}$$

Based on the DSC design (20) and (21), differentiating S_i^δ yields

$$\dot{S}_i^\delta = -\frac{\tilde{\delta}_i^2}{\tau_i}. \tag{103}$$

Collecting theoretical analysis set forth, the following theorem concludes the stability of the tracking and filter errors.

Theorem 4. Let all condition in Theorem 3 hold. The tracking errors z_i for $i = 1, \dots, n$ and the DSC filter errors $\tilde{\delta}_i$ for $i = 1, \dots, n-1$ with bounded initial values, converge to a small neighborhood of the origin, that is, Problem 1 is solved.

Proof. Firstly, consider S_i in (102). Based on (23) and Assumption 3, (102) further becomes

$$\begin{aligned}
\dot{S}_i &\leq -\frac{\kappa_i^2 g_i^2}{2r_i} z_i^2 - \kappa_i z_i \left[g_i \left(\tilde{\theta}_i^A \right)^\top \psi_i^A + k_i \left(\tilde{\theta}_i^D \right)^\top \psi_i^D \right] \\
&\quad + \kappa_i |z_i| \left(\bar{f} + \bar{\ell} + \bar{g} \|\theta_i^{A*}\| \bar{\psi}_A + \bar{k} \|\theta_i^{D*}\| \bar{\psi}_D \right) \\
&\quad + \kappa_i g_i z_i z_{i+1} + \kappa_i g_i z_i \tilde{\delta}_i.
\end{aligned} \tag{104}$$

Applying the Young's inequality on the terms $\kappa_i g_i z_i z_{i+1}$ and $\kappa_i g_i z_i \tilde{\delta}_i$ yields

$$\kappa_i g_i z_i z_{i+1} \leq \frac{\kappa_i \bar{g}}{2} z_i^2 + \frac{\kappa_i \bar{g}}{2} z_{i+1}^2, \quad \kappa_i g_i z_i \tilde{\delta}_i \leq \frac{\kappa_i \bar{g}}{2} z_i^2 + \frac{\kappa_i \bar{g}}{2} \tilde{\delta}_i^2. \tag{105}$$

Substituting (105) into (104) gives

$$\begin{aligned}
\dot{S}_i &\leq -\left(\frac{\kappa_i^2 g_i^2}{2r_i} - \kappa_i \bar{g} \right) z_i^2 - \kappa_i z_i \left[g_i \left(\tilde{\theta}_i^A \right)^\top \psi_i^A + k_i \left(\tilde{\theta}_i^D \right)^\top \psi_i^D \right] \\
&\quad + \kappa_i |z_i| \left(\bar{f} + \bar{\ell} + \bar{g} \|\theta_i^{A*}\| \bar{\psi}_A + \bar{k} \|\theta_i^{D*}\| \bar{\psi}_D \right) \\
&\quad + \frac{\kappa_i \bar{g}}{2} z_{i+1}^2 + \frac{\kappa_i \bar{g}}{2} \tilde{\delta}_i^2.
\end{aligned}$$

Consider the time $t \geq \max \{T_i^A, T_i^D\}$. Then, with Theorem 3, one has $\|\tilde{\theta}_i^A\| \leq \Theta_i^A$ and $\|\tilde{\theta}_i^D\| \leq \Theta_i^D$, which indicates $\dot{S}_i$ can be expressed as

$$\begin{aligned}
\dot{S}_i &\leq -\left(\frac{\kappa_i^2 g_i^2}{2r_i} - \kappa_i \bar{g} \right) z_i^2 \\
&\quad + \kappa_i |z_i| \underbrace{\left(\bar{g} \|\theta_i^{A*}\| \bar{\psi}_A + \bar{k} \|\theta_i^{D*}\| \bar{\psi}_D + \bar{f} + \bar{\ell} \right)}_{\triangleq \beta_i^z} \\
&\quad + \frac{\kappa_i \bar{g}}{2} z_{i+1}^2 + \frac{\kappa_i \bar{g}}{2} \tilde{\delta}_i^2.
\end{aligned} \tag{106}$$

With the Young's inequality $\kappa_i |z_i| \beta_i^z \leq \beta_i^z \frac{z_i^2}{2} + \frac{1}{2} \beta_i^z \kappa_i^2$, (106) further satisfies

$$\dot{S}_i \leq -\left(\frac{\kappa_i^2 g_i^2}{2r_i} - \kappa_i \bar{g} - \frac{\beta_i^z}{2} \right) z_i^2 + \frac{\kappa_i \bar{g}}{2} z_{i+1}^2 + \frac{\kappa_i \bar{g}}{2} \tilde{\delta}_i^2 + \frac{1}{2} \beta_i^z \kappa_i^2. \tag{107}$$

Collecting the results in (103) and (107), the derivative of S satisfies

$$\begin{aligned}
\dot{S} &= \sum_{i=1}^n \dot{S}_i + \sum_{i=1}^{n-1} \dot{S}_i^\delta \\
&\leq -\left(\frac{\kappa_1^2 g_1^2}{2r_1} - \kappa_1 \bar{g} - \frac{\beta_1^z}{2} \right) z_1^2 - \sum_{i=2}^{n-1} \left(\frac{\kappa_i^2 g_i^2}{2r_i} - \frac{3}{2} \kappa_i \bar{g} - \frac{\beta_i^z}{2} \right) z_i^2 \\
&\quad - \left(\frac{\kappa_n^2 g_n^2}{2r_n} - \frac{1}{2} \kappa_n \bar{g} - \frac{\beta_n^z}{2} \right) z_n^2 - \sum_{i=1}^{n-1} \left(\frac{1}{\tau_i} - \frac{\kappa_i \bar{g}}{2} \right) \tilde{\delta}_i^2 + \frac{1}{2} \sum_{i=1}^n \beta_i^z \kappa_i^2.
\end{aligned} \tag{108}$$

Define

$$\begin{aligned}
\varpi_1 &\triangleq 2 \min \left\{ \kappa_1 \left(\frac{\kappa_1^2 g_1^2}{2r_1} - \kappa_1 \bar{g} - \frac{\beta_1^z}{2} \right), \right. \\
&\quad \left. \kappa_i \left(\frac{\kappa_i^2 g_i^2}{2r_i} - \frac{3}{2} \kappa_i \bar{g} - \frac{\beta_i^z}{2} \right) \right|_{i=2}^{n-1}, \\
&\quad \left. \kappa_n \left(\frac{\kappa_n^2 g_n^2}{2r_n} - \frac{1}{2} \kappa_n \bar{g} - \frac{\beta_n^z}{2} \right), \left(\frac{1}{\tau_i} - \frac{\kappa_i \bar{g}}{2} \right) \right|_{i=1}^{n-1} \right\}, \tag{109}
\end{aligned}$$

$$\varpi_2 \triangleq \frac{1}{2} \sum_{i=1}^n \beta_i^z \kappa_i^2. \tag{110}$$

Then, (108) can be further expressed as

$$\dot{S}(\bar{z}, \bar{\delta}) \leq -\varpi_1 S(\bar{z}, \bar{\delta}) + \varpi_2. \tag{111}$$

Applying the comparison Lemma [2], one can obtain the following

$$\begin{aligned}
S(\bar{z}(t), \bar{\delta}(t)) &\leq e^{-\varpi_1 t} S(\bar{z}(0), \bar{\delta}(0)) + \frac{\varpi_2}{\varpi_1} (1 - e^{-\varpi_1 t}) \\
&\leq S(\bar{z}(0), \bar{\delta}(0)) + \frac{\varpi_2}{\varpi_1},
\end{aligned} \tag{112}$$

which guarantees that after $t \geq \max \{T_i^A, T_i^D\}$, the tracking and DSC filter errors converge exponentially fast to the residual sets:

$$z_i(t) \in \Omega^z \triangleq \left\{ z_i \in \mathbb{R} \mid |z_i| \leq \sqrt{2 \kappa_i \left(P(\bar{z}(0), \bar{\delta}(0)) + \frac{\varpi_2}{\varpi_1} \right)} \right\}$$

$$\text{for } i = 1, \dots, n \quad (113)$$

$$\tilde{\delta}_i(t) \in \Omega^\delta \triangleq \left\{ \tilde{\delta}_i \in \mathbb{R} \mid |\tilde{\delta}_i| \leq \sqrt{2 \left(P(\bar{z}(0), \bar{\delta}(0)) + \frac{\varpi_2}{\varpi_1} \right)} \right\}$$

$$\text{for } i = 1, \dots, n-1 \quad (114)$$

Based on (109), by choosing the parameters κ_i , τ_i , and r_i such that ϖ_1 is sufficiently large, above residual sets Ω^z and Ω^δ can be made as small neighborhoods of the origin contained by the compact set Ω_i . $\square$

Up to this point, the adaptive weight errors, tracking errors, and DSC filter errors have been shown to be UUB, with the convergence and boundedness of the input pair $\{\hat{u}_i^A, \hat{d}_i^D\}$ being verified. Therefore, all signals in the closed-loop system are ensured to be bounded. Moreover, under the proposed Actor-Critic-Disturbance learning structure in Section 4.3, the tracking performance and stability are guaranteed even in the presence of the approximate worst-case disturbance inputs, thereby confirming the robustness of the presented scheme.

5 | Simulation Studies

Two illustrative cases are investigated in this section to validate the robustness and effectiveness of the Actor-Critic-Disturbance learning algorithm presented in Section 4.3. Specifically, for function approximation the first example employs PBFs for simplicity, whereas the second utilizes the RBFs.

5.1 | Case 1: Numerical Example

Consider the third-order strict-feedback nonlinear system in the form of

$$\begin{aligned} \dot{x}_1 &= 0.1x_1^2 + x_2, \\ \dot{x}_2 &= 0.1x_1x_2 - 0.2x_1 + (1 + x_1^2)x_3, \\ \dot{x}_3 &= x_1x_3 + 10u + 7d, \end{aligned} \quad (115)$$

where x_1 is the system output to be controlled to track the desired reference signal $x_d(t) = \sin(t) + \cos(3t)$. u the minimizing player (Actor) and d the maximizing player (Disturbance) constitute the real input tuple. Respectively, the tracking errors z_i , DSC filters and augmented states $\mathcal{X}_i$ are denoted in (18), (20), and (25). For function approximation, the PBF for $p \in \{A, C, D\}$, $i = 1, 2, 3$ is determined as

$$\psi_i^p(\mathcal{X}_i) = \psi_i^p([z_i, \delta_{i-1}]) = [z_i^2, z_i\delta_{i-1}, \delta_{i-1}^2]^\top. \quad (116)$$

Given (116), each input pair $\{\hat{u}_i^A, \hat{d}_i^D\}$ are designed according to (58) and (59) with the coefficients $\kappa_i = 32$. The Actor-Critic-Disturbance learning laws are specified in (71), (66), and (76), with learning rates $\alpha_1^C = 0.5$, $\alpha_1^A = \alpha_1^D = 0.01$; $\alpha_2^C = 0.8$, $\alpha_2^A = \alpha_2^D = 0.03$; $\alpha_3^C = 1.2$, $\alpha_3^A = \alpha_3^D = 0.15$. The performance output (27) of each subsystem is determined by $q_i = r_i = 4$. In the performance index (28), the discount factor is chosen as $\eta = 0.1$. For initialization, the system states x_i , the DSC internal states δ_1 and δ_2 , together with the adaptive weights $\hat{\theta}_{i,j}^p$ for $j = 1, 2, 3$, are all set randomly from $[-1, 1]$. The time constants in DSC filters are $\tau_1 = 0.1$ and $\tau_2 = 0.01$. The simulation results corresponding to above settings are illustrated in Figures 2–4.

In Figure 2, despite the presence of approximate worst-case disturbances, the states of the system (115) are shown to track the corresponding trajectories, which confirms the analysis in Theorem 4. Figure 3 depicts the adaptive evolution of the Actor and Disturbance weights. Notice that the third component of each Actor and Disturbance weight exhibit obvious oscillation, while the other two components show smoother transient behavior. This is because the selection of the PBF in (116), such that the third components $\hat{\theta}_{i,3}^A$ and $\hat{\theta}_{i,3}^D$ capture the time-varying property of the reference signals δ_{i-1} , for $i = 1, 2, 3$. With $\hat{\theta}_{i,3}^A$ and $\hat{\theta}_{i,3}^D$ fluctuating within bounded ranges, satisfactory tracking performance can still be achieved as observed in Figure 2, which is consistent with the theoretical analysis in Theorems 2 and 3. The similar oscillatory pattern is observed in Figure 4 for both the control and the disturbance input, and the correspondence in their fluctuation trends reflects the tracking objective. Collectively, Figures 2–4 demonstrate that the system maintains satisfactory tracking performance despite the presence of the worst-case disturbances.

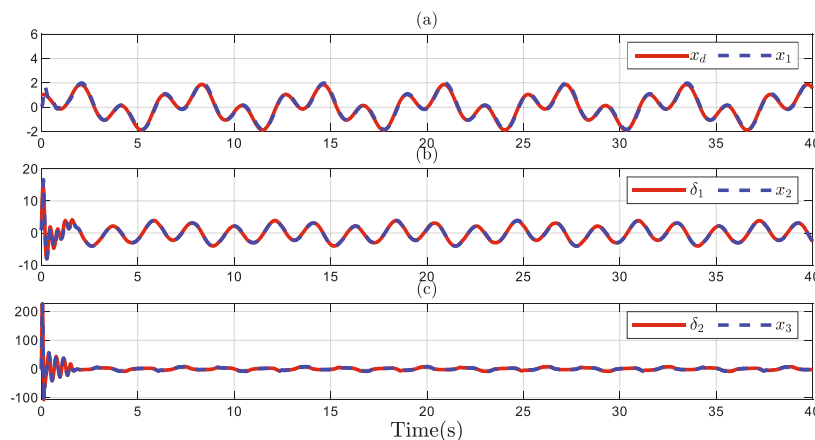


FIGURE 2 | Trajectories of the states in system (115) with $x_d(t) = \sin(t) + \cos(3t)$.

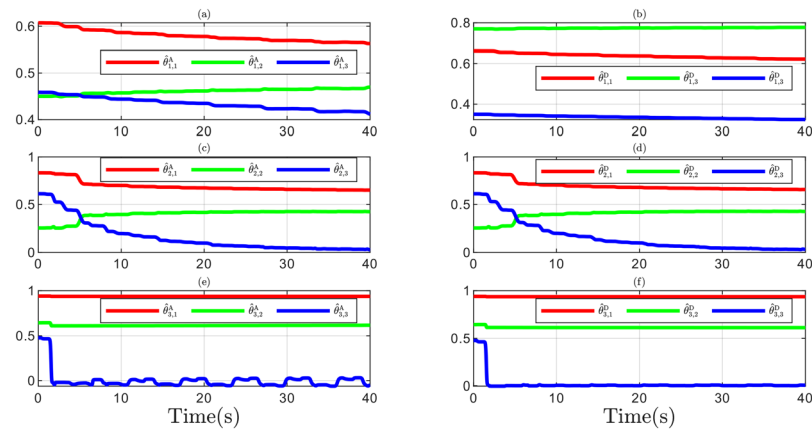


FIGURE 3 | Evolution of the adaptive weights in system (115).

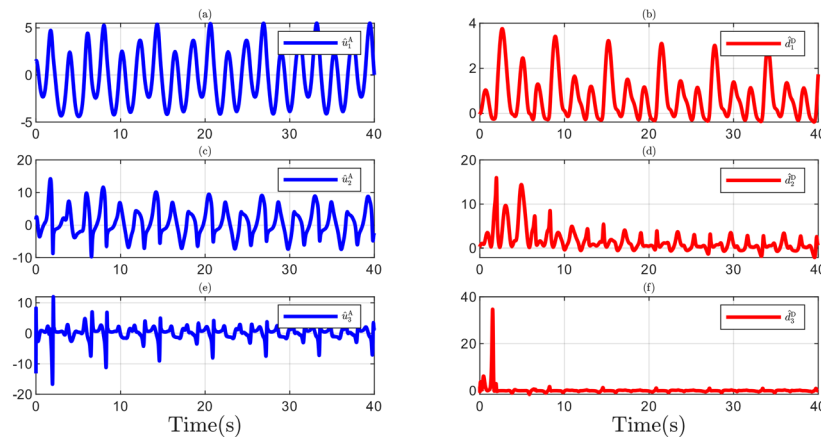


FIGURE 4 | Evolution of the input signals in system (115).

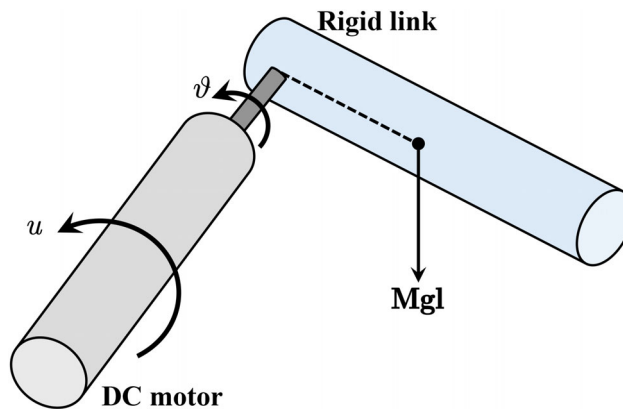


FIGURE 5 | Single link robot arm.

5.2 | Case 2: Practical Example

From an application viewpoint, the proposed framework is particularly suitable for engineering systems where the dominant uncertainty can be modeled as a finite-energy disturbance. Typical examples include robotic manipulators subject to bounded-energy unmodeled dynamics and impacts, ground vehicles exposed to road excitations, and power-electronic systems (e.g., DC microgrids) undergoing bounded-energy load

fluctuations. In such scenarios, modeling the disturbance as an adversarial player with an $\mathcal{L}_2$ -energy budget provides a principled way to quantify disturbance attenuation via an H_∞ objective.

The effectiveness of the developed design is further illustrated by single link robot arm [56] system as shown in Figure 5, which can be expressed in the Lagrangian form:

$$\mathbf{J}\ddot{\theta} + \mathbf{B}\dot{\theta} + \mathbf{M}\mathbf{g}\mathbf{l}\sin\theta = u(t) + d(t), \quad (117)$$

with input pair $\{u, d\}$. The model parameters are specified in Table 1. Define the states $x_1(t) \triangleq \theta(t)$, $x_2(t) \triangleq \dot{\theta}(t)$, and $x(t) \triangleq [x_1(t), x_2(t)]^\top$. Then, the state-space expression of the robot arm (117) is given by

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -9.8 \sin x_1 - 2x_2 + u + d. \end{aligned} \quad (118)$$

TABLE 1 | System parameters of the one-link manipulator.

Symbol	Physical meaning	Value
θ	Angle of the link	—
$\dot{\theta}$	Angular velocity of the link	—
$\mathbf{g}$	Gravitational acceleration	9.8 m/s ²
$\mathbf{B}$	Overall damping coefficient	2 kg · m/s
$\mathbf{M}$	Total mass of the link	1 kg
$\mathbf{J}$	Rotational inertia of the motor	1 kg · m ²
$\mathbf{l}$	Distance from joint axis to center of mass	1 m

In this case, the desired trajectory to be tracked by the system output x_1 is selected as $x_d(t) = \sin(2t)$. The coordination transformation, system augmentation, and DSC filter design are set forth in Section 3. For Actor-Critic-Disturbance representation, we utilize the Gaussian functions in (15) for RBFNNs. Let $\psi_i^p(\mathcal{X}_i) = \psi_i^p([z_i, \delta_{i-1}])$ be composed of 9 nodes. Then, with the all the widths being 1, the centers are determined to be spaced evenly in the region $[-3, 0, 3] \times [-3, 0, 3]$, $[-2, 0, 2] \times [-2, 0, 2]$, and $[-1, 0, 1] \times [-1, 0, 1]$ for each Critic, Actor, and Disturbance NN, respectively. Additionally, the initial adaptive weights are all chosen randomly from $[-1, 1]$. The initial state is $x(0) = [0.1, -0.1]^\top$. For the DSC filter, we set $\delta_1(0) = 0.32$ and $\tau_1 = 0.003$.

For the performance parameters, we set $r_1 = q_1 = 0.1$, and $r_2 = q_2 = 0.01$, with the discount factor $\eta = 0.1$. The learning rates are given by $\alpha_i^C = 0.1$, $\alpha_i^A = \alpha_i^D = 0.01$. Finally, the control coefficients κ_i in (50) are chosen as $\kappa_i = 50$, compared with the cases for $\kappa_i = 30$ and $\kappa_i = 10$. Then, the results are depicted by Figures 6–10.

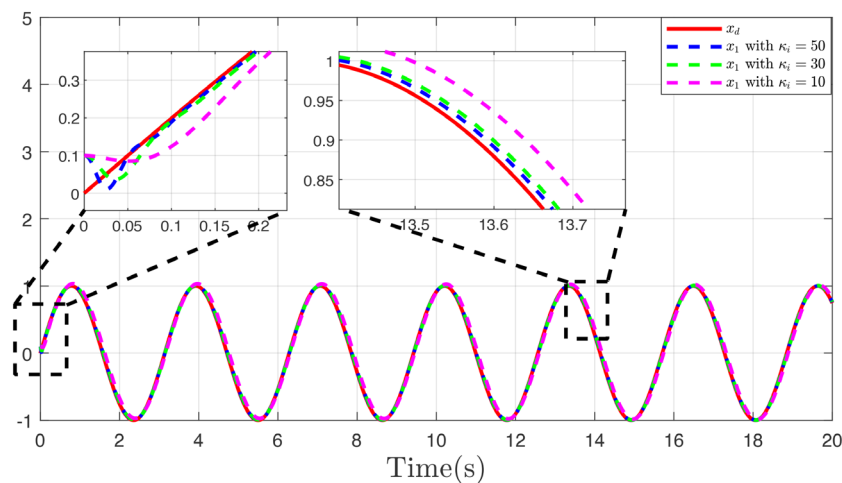


FIGURE 6 | Tracking performance of the angular x_1 in the robot system (117) under $\kappa_i = 10, 30, 50$, with $x_d(t) = \sin(2t)$.

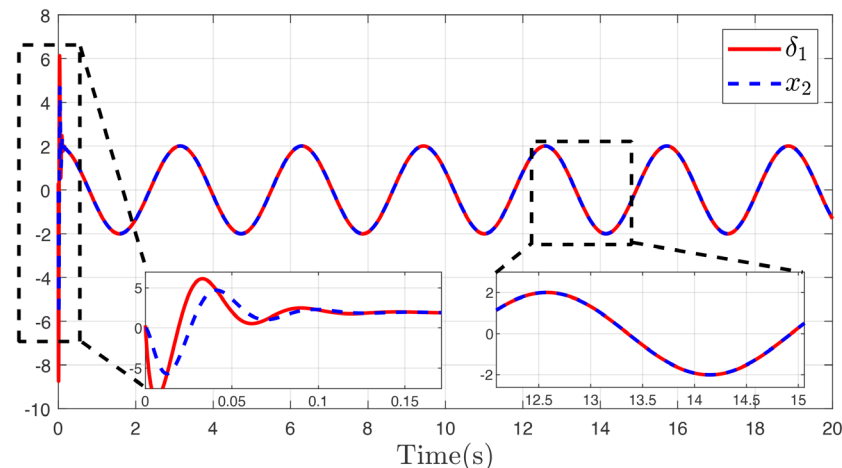


FIGURE 7 | Trajectory of the angular velocity x_2 in the robot system (117).

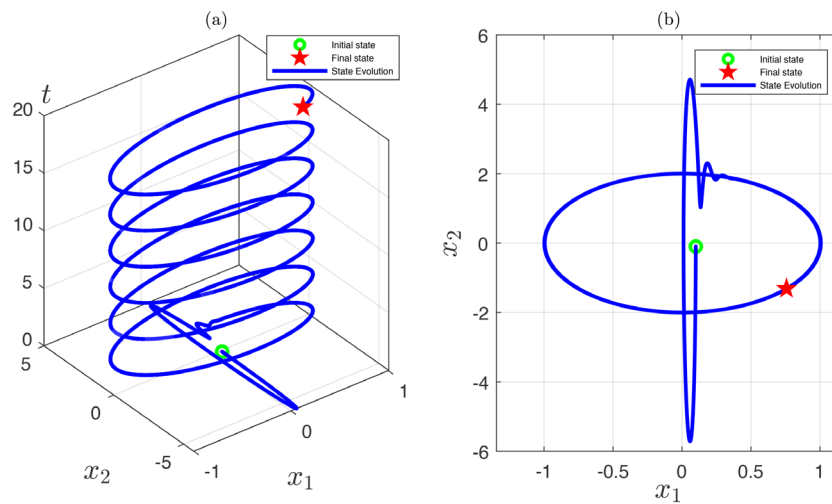


FIGURE 8 | Phase portrait of the state trajectories $[x_1(t), x_2(t)]$. (a) The phase portrait with time evolution. (b) The 2-dimensional phase plot.

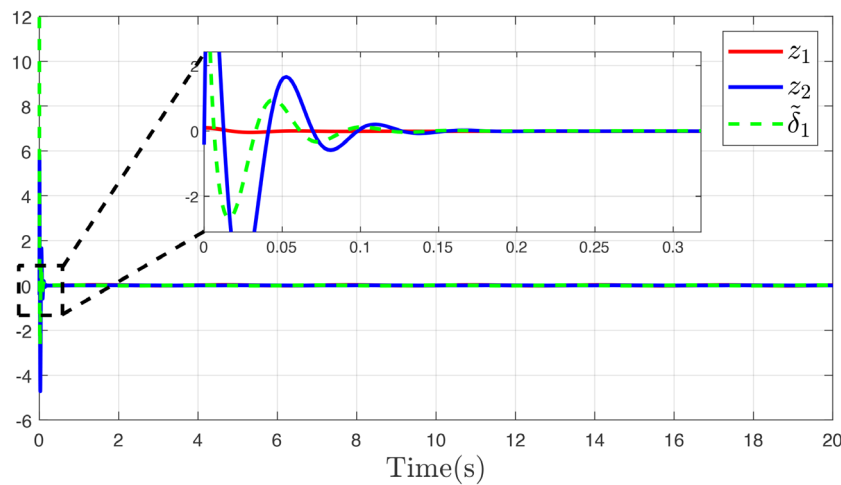


FIGURE 9 | Evolution of the tracking errors z_i and DSC filter error $\tilde{\delta}_1$ in the robot system (117).

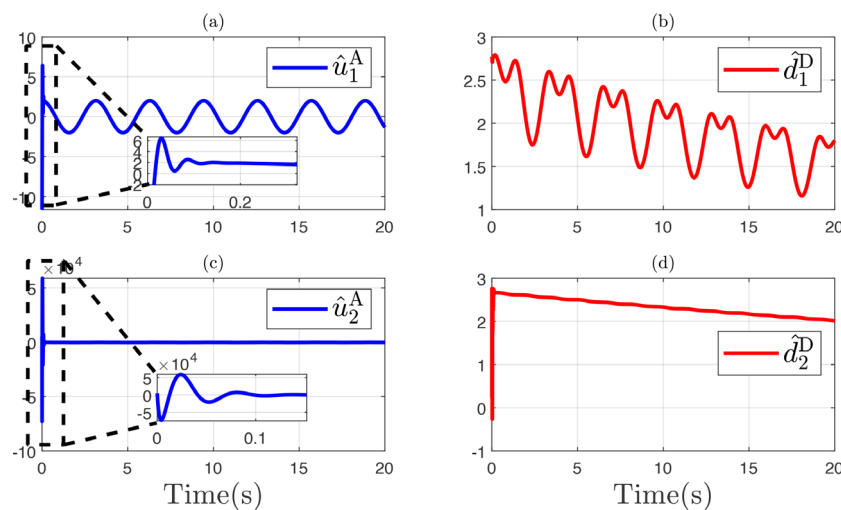


FIGURE 10 | Evolution of the input signals in the robot system (117).

Figure 6 compares the angular trajectory x_i with the reference $x_d(t) = \sin(2t)$ under different choices of κ_i . All simulated responses exhibit desired tracking performance, while the zoomed-in plots highlight the differences in transient behavior. Specifically, larger κ_i values lead to reduced deviation from the reference, which confirms the discussion in Theorem 4 and reflects the role of the quadratic term S_i in (50) in shaping the transient performance. This behavior is consistent with the stability result in Theorem 2, where the Critic learning ensures a bounded approximation of the value function, thereby enabling the Actor to generate control inputs that stabilize the tracking process despite parameter variation.

Figure 7 depicts the evolution of the angular velocity x_2 and its reference, that is, the DSC internal state δ_1 . The trajectories overlap closely within a transient phase of 0.15 s, with initial overshoots reflecting the dynamics of filter compensation, which can be observed in the zoomed-in plots. The bounded evolution of both signals illustrates the capability of the DSC structure to smooth virtual control inputs, ensuring that the control effort remains implementable. The observed behavior links with Theorem 4 and the discussions below Theorem 4, which guarantees that the tracking errors and filter states converge to a small neighborhood of the origin.

The phase portraits in Figure 8 give a more intuitive view of how x_1 and x_2 evolve together. While Figures 6 and 7 separately show that x_1 tracks the reference and x_2 converges toward the filter output, the spiral trajectories here reveal their coupled dynamics: after transient oscillations, the states settle onto a bounded orbit. This joint convergence illustrates the effect of the Actor-Critic-Disturbance learning, consistent with the boundedness and stability results in Theorem 4.

Figure 9 presents the evolution of the tracking errors z_1, z_2 and the DSC filter error $\tilde{\delta}_1$. After sharp initial oscillations (see the zoomed-in plot), all error trajectories diminish rapidly and remain confined within 0.2 s. This transient behavior is attributed to the adaptive learning process of the Actor-Critic-Disturbance networks in Section 4.3, which gradually refine their weight estimates and thereby reducing approximation errors. The convergence of these signals is in line with the conclusions of Theorems 2 to 4, which together ensure that adaptive weight errors remain UUB and that the tracking errors approach neighborhoods arbitrarily close to the origin.

The time evolution of the control and disturbance signals are illustrated in Figure 10. The virtual/real controller $\hat{u}_i^A$ in Figure 10a,c exhibit bounded oscillations, while the disturbance input $\hat{d}_i^D$ in Figure 10b,d adapts dynamically to counter the control effects of $\hat{u}_i^A$, which forms the ZSG as discussed in Section 3. Collectively, these curves highlight the coordinated role of the Actor-Critic-Disturbance networks: the Critic approximates the value function to guarantee boundedness (Theorem 2), while the Actor-Disturbance pair converges to an approximate Nash solution (Theorem 3), and the overall closed-loop stability follows from Theorem 4.

6 | Conclusion

This article proposes a robust adaptive H_∞ tracking control scheme for nonlinear strict-feedback systems by reformulating the backstepping design into a ZSG, which directly decouples the tracking and DSC filter errors and thus simplifies analysis. The Actor-Critic-Disturbance learning framework avoids solving the HJI equation directly by decomposing the value function and approximating only the nonlinear component with RBFNNs, and the tracking performance is guaranteed by the quadratic part. The implementation of experience replay strategy with the SE condition ensures practical online adaptation, thereby relaxing the conventional requirement of PE. Theoretical analysis and simulations confirm that the proposed approach ensures the tracking performance with tracking errors being UUB even in the presence of worst-case disturbance, highlighting its potential as a practical and innovative framework for robust adaptive control. Future works will extend the results to event/self triggered control and multi-agent systems.

Notations

$\mathbb{R}$	the set of real numbers;
$\mathbb{R}^+$	the set of positive real numbers;
$\mathbb{R}^n$	the n -dimensional Euclidean space;
$\mathbb{R}^{m \times n}$	the set of $m \times n$ -dimensional real matrices;
$\mathbf{I}_n$	the n -dimensional identity matrix;
$\mathbf{0}_{m \times n}$	the $m \times n$ -dimensional zero matrix;
$\lambda_{\min}(A)$	the minimum eigenvalue of matrix A ;
$A^\top$	the transpose of a vector or matrix A ;
$\nabla^\top f$	the transposed gradient of function f ;
$\mathcal{L}_2$	the space of functions with finite energy;
$\ x\ $	$\sqrt{x^\top x}$, the Euclidean norm of vector x ;
$\ x\ _{\mathcal{L}_2}$	$\sqrt{\int_0^\infty x^\top(t)x(t) dt}$, the $\mathcal{L}_2$ norm of vector signal x ;
$\ x\ _{\mathcal{L}_\infty}$	$\max_i x_i $, the $\mathcal{L}_\infty$ norm for a vector $x = [x_1, \dots, x_i, \dots, x_n]^\top$.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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